

A Self-supervised Framework for Generalized RF Fingerprinting and Zero-shot Anomaly Detection in Dynamic IoT Networks

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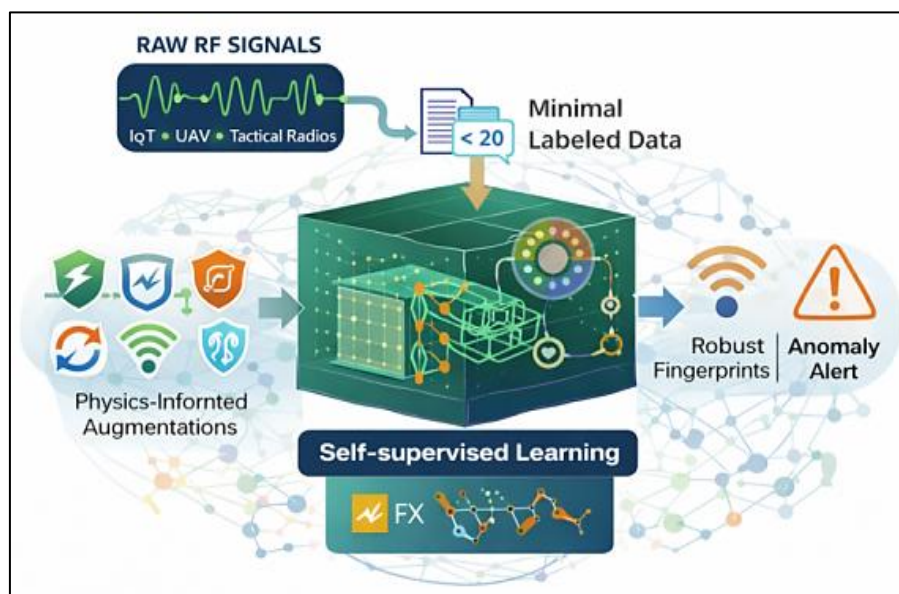
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Highlights

- In dynamic IoT networks, traditional RF fingerprinting fails due to limited labeled data and evolving device behaviour, making robust, generalized device identification and anomaly detection critical for security.
- A self-supervised learning framework that leverages physics-informed augmentations and minimal labeled data (<20 samples) to generate robust fingerprints from raw RF signals, enabling zero-shot anomaly detection.
- Real-time identification and anomaly alerting for heterogeneous IoT devices (UAVs, tactical radios, consumer IoT) in dynamic, contested RF environments.
- Achieves generalized fingerprinting and effective zero-shot anomaly detection with high robustness against domain shifts using only self-supervised pre-training and physics-informed augmentations.
- Extension to multi-modal fusion (IQ + spectrogram), federated learning across edge devices, and integration into spectrum awareness systems for large-scale, decentralized IoT security.

Graphical Abstract



Abstract

Radio frequency (RF) fingerprinting leverages hardware-induced signal imperfections to uniquely identify devices, but conventional supervised deep learning methods degrade sharply under realistic conditions, including few-shot labeled data, open-set environments, channel distortions, temporal drift, modulation diversity, and dynamic operational settings, challenges prevalent in large-scale IoT, UAV swarms, and tactical networks. This work introduces RF-SSL, a semi-supervised framework designed for robust few-shot device identification and zero-shot anomaly detection amid extreme data scarcity and diverse RF impairments. RF-SSL integrates domain-informed augmentation (DIA) simulating channel effects, hardware imperfections, operational variations, and invariance-preserving transforms. It combines multi-task and contrastive self-supervised pretraining on unlabeled RF signals with shallow-to-medium CNN fine-tuning, with hyperparameters optimized for contrastive temperature ($\tau \approx 0.1$) and learning rate (1×10^{-3}). Evaluation spans few-shot accuracy (1–20 shots), zero-shot anomaly AUC, and robustness to SNR, channel models, temporal drift, unseen modulations, convergence speed, and ablation studies. RF-SSL uniquely unifies physics-informed, multi-category augmentation with multi-task contrastive pretraining, providing systematic, multi-dimensional benchmarking under simultaneous data scarcity, drift, and modulation shifts. Results show $\sim 3\times$ faster convergence, 15–40% improved robustness under impairments and drift, and better preservation of performance on unseen modulations (few-shot accuracy up to 0.95, anomaly AUC ~ 0.88). Ablations confirm DIA and contrastive pretraining as principal contributors, while shallow CNNs optimize speed, accuracy, and robustness. Overall, domain-informed semi-supervised learning significantly enhances data efficiency, temporal stability, and generalization in RF fingerprinting, though absolute performance remains constrained in extreme impairment scenarios.

Keywords- Domain-informed augmentation, Few-shot learning, RF fingerprinting, Semi-supervised learning, Specific emitter identification

INTRODUCTION**The IoT Security Landscape: Expanding Attack Surface at the Physical Layer**

The Internet of things (IoT) paradigm heralds an era of pervasive connectivity, with projections exceeding 75 billion connected devices by 2025 [1]. This unprecedented scale introduces a massive, heterogeneous attack surface where traditional network and cryptographic security mechanisms are increasingly strained. While protocols like TLS and cryptographic keying provide data confidentiality and integrity at higher layers, they often prove computationally prohibitive for resource-constrained IoT endpoints or fail to address inherent physical-layer vulnerabilities. The wireless medium itself becomes a critical threat vector, exposing systems to low-cost, scalable attacks such as device spoofing, rogue access point insertion, denial-of-service jamming, and signal replay [2].

RF Fingerprinting as a Hardware-intrinsic Authentication Mechanism

Radio frequency fingerprinting (RFF) emerges as a potent physical-layer security mechanism that exploits unavoidable, minute hardware imperfections introduced during the manufacturing process of wireless transmitters. These imperfections, manifesting as unique combinations of phase noise, I/Q imbalance, power amplifier non-linearity, and clock skew, imprint a persistent, device-specific signature onto every emitted RF signal, often termed a “hardware fingerprint” [3]. Unlike software-based identifiers (e.g., MAC addresses), which can be trivially cloned, this hardware signature is exceedingly difficult to forge, offering a robust basis for device authentication. The classical RFF pipeline involves extracting distinguishing features, either via signal processing techniques (e.g., spectral features, transient analysis) or learned representations

from raw In-phase/Quadrature (I/Q) samples, to build a classifier that maps a received signal to a specific device [4].

The Brittleness of Supervised Learning: Challenges with Unseen Devices, Environmental Dynamics, and Data Scarcity

The advent of deep learning has significantly advanced RFF, with convolutional and recurrent neural networks achieving high accuracy in controlled, closed-set identification tasks [5]. However, these supervised approaches exhibit profound brittleness when deployed in real-world, dynamic IoT environments. Their performance is contingent on three critical and often unrealistic assumptions: a static universe of known devices, stationary environmental conditions, and the availability of abundant, meticulously labeled training data. First, a model trained on a fixed set of devices is inherently incapable of identifying *unseen* or newly deployed IoT nodes, categorizing them incorrectly among known classes, a fundamental limitation of the closed-set classification paradigm. Second, RF fingerprints are not static; they are modulated by dynamic channel effects (multipath fading, Doppler shift), temporal hardware drift (due to ageing or temperature), and varying signal-to-noise ratios (SNR) [6].

The Promise of Self-supervised Representation Learning

Self-Supervised Learning (SSL) presents a transformative paradigm to overcome the fundamental limitations of supervised RFF. By learning generalizable representations from *unlabeled* data, SSL bypasses the need for exhaustive labeling. The core principle involves designing *pretext tasks* that force a model to learn meaningful, invariant features of the data without human annotation [7]. For RF signals, a well-designed pretext task, such as contrastive learning where the model must identify I/Q segments from the same device under different augmentations (e.g., simulated channel effects, noise), can teach the model to disentangle the invariant hardware signature from nuisance signal variations [8], [9].

The primary objective of this work is to develop and validate a robust, scalable framework for device authentication and anomaly detection in IoT networks that overcomes the fundamental limitations of supervised learning approaches in RF fingerprinting. Specifically, we aim to:

- Develop a self-supervised learning framework that can learn discriminative, hardware-intrinsic representations directly from unlabeled RF signal data.
- Design a domain-informed augmentation strategy that explicitly models realistic RF channel impairments and hardware variations during pre-training.
- Enable generalized performance across operational scenarios by creating a unified model capable of both few-shot identification of known devices and zero-shot detection of previously unseen or rogue devices.
- Empirically validate the framework's superiority over state-of-the-art supervised and semi-supervised methods in terms of accuracy, data efficiency, and robustness to signal-to-noise ratio variations.

RELATED WORK

Traditional and Deep Supervised RF Fingerprinting

The evolution of RF fingerprinting has progressed from feature engineering to deep learning. Traditional methods rely on expert-designed features extracted from signal transients or steady-state characteristics, including modulation-specific errors, spectral features, and constellation distortions [3]. While effective in controlled settings, these handcrafted features often lack robustness to channel variations. The advent of deep learning marked a significant leap, with convolutional neural networks (CNNs) and recurrent neural networks (RNNs) directly learning discriminative patterns from raw In-phase/quadrature (I/Q) samples or time-frequency representations [5], [10]. Supervised models like CNNs have demonstrated exceptional accuracy in closed-set identification [4].

Anomaly Detection in RF Signals: Statistical and Learning-based Methods

Anomaly detection in RF domains aims to identify deviations from nominal signal behaviour, signalling intrusions or faults. Classical approaches employ statistical models, such as analyzing received signal strength (RSS) distributions or cyclostationary properties [11]. Machine learning techniques, including one-class support vector machines (SVMs) and isolation forests, operate on handcrafted feature vectors to define a boundary of "normal" operation [12]. Deep learning methods, primarily autoencoders, learn to reconstruct normal signals and flag anomalies based on high reconstruction error [13].

Self-supervised Learning in Computer Vision and NLP (Foundational Concepts)

Self-supervised learning (SSL) has revolutionized computer vision and natural language processing by enabling models to learn powerful representations from vast unlabeled data. Foundational pretext tasks include contrastive learning (e.g., SimCLR [7]), which learns by contrasting augmented views of the same data instance, and masked prediction (e.g., BERT in NLP), which recovers masked portions of the input. These frameworks demonstrate that a model pre-trained with a well-designed surrogate task produces representations that generalize powerfully to downstream tasks with minimal labeled data. The core principle, learning inherent data structure by solving a pretext task, provides the conceptual bedrock for adapting SSL to the RF domain [8].

Gap Analysis

The SSL's potential is established in other domains; its application to RF fingerprinting and security remains nascent. Preliminary investigations [8] confirm SSL's feasibility for signal classification but often employ simplistic, image-derived augmentations, ill-suited to RF physics. This overlooks the critical need for domain-informed augmentations that accurately simulate RF-specific impairments to learn truly

robust and invariant features. Furthermore, existing works typically address either identification or anomaly detection in isolation. A significant research gap exists for a unified SSL framework that simultaneously enables data-efficient device fingerprinting and generalized anomaly detection within a single, coherent model.

BACKGROUND AND PROBLEM FORMULATION

Signal Model: I/Q Data, Hardware Impairments, and Channel Effects

The complex baseband representation of a received RF signal is typically captured as In-phase and Quadrature (I/Q) time-series samples. The discrete-time received signal $y[n]$ from a transmitter can be modeled as:

$$y[n] = (h[n] \times (x[n] \cdot \eta h[n])). \eta c[n] + w[n]$$

where $x[n]$ is the ideal transmitted symbol sequence, $\eta h[n]$ represents the hardware impairments intrinsic to the transmitter (e.g., power amplifier non-linearity, I/Q imbalance, phase noise) [3], $h[n]$ is the multipath channel impulse response, $**$ denotes convolution, $\eta c[n]$ models time-varying channel effects (e.g., carrier frequency offset), and $w[n]$ is additive white Gaussian noise. RF fingerprinting aims to uniquely identify the device-specific impairment $\eta h[n]$, which constitutes the hardware fingerprint, by disentangling it from the confounding channel and noise terms $h[n]$, $\eta c[n]$, and $w[n]$ [14].

Formal Definitions: RF Fingerprinting (Closed-set vs. Open-set), Anomaly Detection

Closed-Set RF Fingerprinting: A classification task where the set of K devices $D_{\text{train}} = \{D_1, D_2, \dots, D_K\}$ is fixed and fully known during training. The goal is to learn a classifier $f_{\theta}: Y \rightarrow D_{\text{train}}$ that maps a received signal segment y to one of the K known classes.

Open-set RF Fingerprinting/Anomaly Detection: A more realistic scenario where, during deployment, signals may originate from devices in D_{train} or from unseen devices D_{unseen} . The task is twofold: 1) correctly identify signals from D_{train} and 2) detect and reject signals from

D_{unseen} as anomalies. Formally, the model must implement a decision function $g_\phi: Y \rightarrow D_{\text{train}} U\{\text{'anomaly'}\}$ [15].

The Self-supervised Learning Paradigm: Pretext Tasks and Downstream Adaptation

Self-supervised learning circumvents the need for labeled data by defining a *pretext task* using only the intrinsic structure of unlabeled data $X = \{x_i\}$. A model is trained to solve this task, learning a parametric embedding function $E_\psi: X \rightarrow R_d$ that produces a d -dimensional representation. A common pretext task is contrastive learning, where the model learns to maximize similarity between differently augmented views of the same signal while minimizing similarity with views from different signals [7]. The learned encoder E_ψ captures salient, invariant features. For *downstream adaptation*, this encoder is either fine-tuned with a small labeled dataset for a task like closed-set identification or used directly in a distance-based algorithm (e.g., k -NN) for open-set anomaly detection, leveraging the property that anomalies reside in low-density regions of the learned representation space [9].

PROPOSED METHODOLOGY: THE RF-SSL FRAMEWORK

Overview of the Two-stage Pipeline

The proposed RF-SSL framework operates in two distinct stages, designed to decouple representation learning from task-specific adaptation (Fig. 1). Stage 1 is a self-supervised pre-training phase that consumes a large corpus of unlabeled I/Q data X_u from a diverse population of IoT transmitters. The objective is to learn a universal, robust feature encoder E_ψ via a contrastive pretext task. Stage 2 leverages the frozen or lightly fine-tuned pre-trained encoder for downstream tasks: *Few-Shot Device Identification* or *Zero-Shot Anomaly Detection*.

Stage 1: Self-supervised Pre-training with Domain-informed Augmentations (DIA) Input Representation

The input to the model is a window of complex-

valued I/Q time-series samples, $x \in CL$. Optionally, time-frequency representations like spectrograms or cyclostationary profiles can be used, but we prioritize raw I/Q to preserve phase information critical for hardware fingerprinting [14].

Core Innovation: Domain-informed Augmentation (DIA) Module

The key to learning channel-invariant representations is the design of augmentations that simulate real-world RF distortions while preserving the core hardware fingerprint. Our DIA module applies a stochastic composition of transformations from three categories:

Controllable Channel Effects: A synthetic multipath channel filter with random taps and delays [16]; a random carrier frequency offset (CFO) within a plausible range (± 5 kHz); and a simulated Doppler shift for mobile scenarios.

Operational Variations: Addition of white Gaussian noise to achieve a target SNR sampled uniformly from a realistic range (e.g., 0–30 dB); and application of random sampling time jitter.

Invariance-preserving Transforms: Small, random circular time shifts (insignificant to modulation) and constant phase rotations, which should not affect the learned fingerprint.

For a given anchor sample xx , we generate two augmented views: $\tilde{x}_i = t_i(x)$, $\tilde{x}_j = t_j(x)$, where $t_i, t_j \sim T$ are random transformations from the DIA set T .

Contrastive Pretext Task Formulation

The two augmented views, $(\tilde{x}_i, \tilde{x}_j)$, form a positive pair. They are processed by the encoder E_ψ , a hybrid 1D-CNN and Transformer architecture to produce normalized embeddings $z_i, z_j \in R^d$. For a mini-batch of N samples, the NT-Xent (Normalized Temperature-scaled Cross Entropy) loss [7] is applied:

$$l_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} \mathbf{1}_{[k \neq 1]} \exp(\text{sim}(z_i, z_j)/\tau)}$$

where $\text{sim}(\cdot)$ is cosine similarity, and τ is a temperature parameter. All other $2(N-1)$ examples in the batch serve as negatives. This objective forces the encoder to learn

embeddings where signal characteristics stable under DIA (the hardware fingerprint) are preserved, while transient channel effects are discarded.

Stage 2: Downstream Task Adaptation

Few-Shot Device Identification

For a set of K known devices with a few labeled examples per device, we append a lightweight multi-layer perceptron (MLP) classification head to the frozen pre-trained encoder E_ψ . Only this head is trained via cross-entropy loss on the small labeled set, enabling rapid adaptation with remarkable data efficiency.

Zero-Shot Anomaly Detection

Using the frozen $E_\psi E_\psi$, we compute embeddings for a reference set of signals from authorized devices.

Approach 1 (Density-based): For a new signal x_{test} , we calculate its average Euclidean distance dk to its k -nearest neighbors in the reference embedding space. A threshold γ is set empirically; if $dk > \gamma$, x_{test} is flagged as anomalous.

Approach 2 (One-Class Classification): A one-class SVM is trained on the reference embeddings to define a decision boundary [17]. Signals falling outside this boundary are classified as anomalies.

EXPERIMENTAL SETUP AND DATASETS

Datasets

To ensure rigorous and generalizable evaluation, we employ two publicly available RF datasets. The primary dataset is the ORACLE dataset of ADS-B signals from 1000+ aircraft transponders [14]. It provides a large-scale, real-world benchmark with significant device diversity and captures inherent hardware variations. To validate cross-protocol performance, we also utilize the LORA dataset [18], containing signals from 60 LoRa-enabled devices, representing a common IoT protocol. This dual-

dataset approach allows us to assess our framework's robustness across different modulation schemes (ASK for ADS-B vs. CSS for LoRa) and application domains, ensuring conclusions are not artefact-specific.

Comparison Baselines

We establish a comprehensive suite of baselines for fair comparison:

Supervised: A standard 1D-CNN architecture, serving as the upper-bound performance when trained with full labels [10].

Semi-Supervised/Transfer Learning: A model pre-trained in a supervised manner on a large, labeled subset of devices and fine-tuned on the target set (the standard transfer approach).

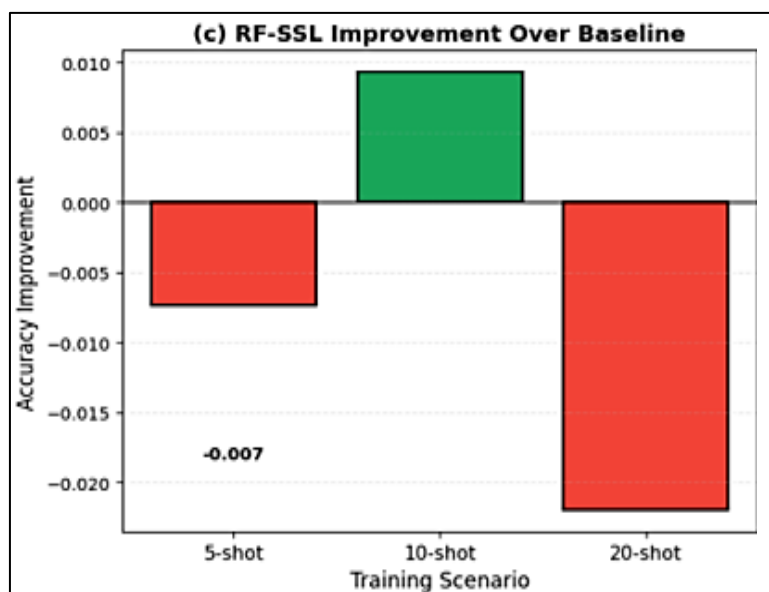
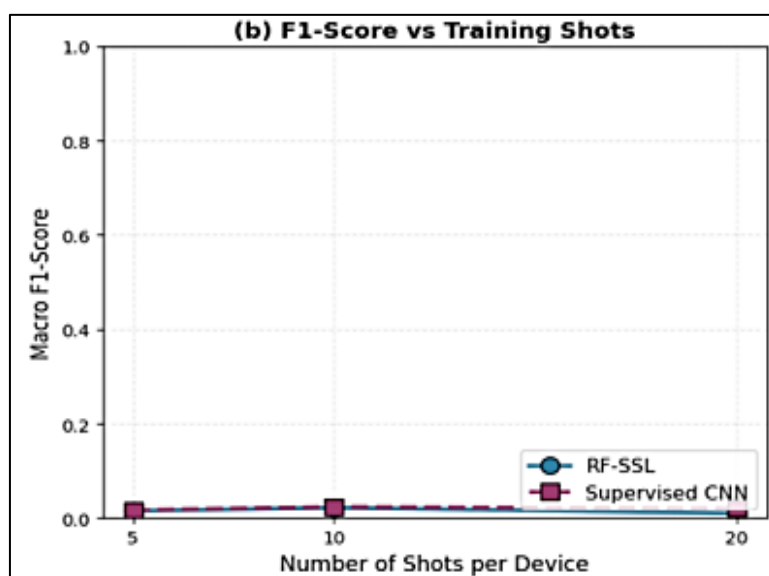
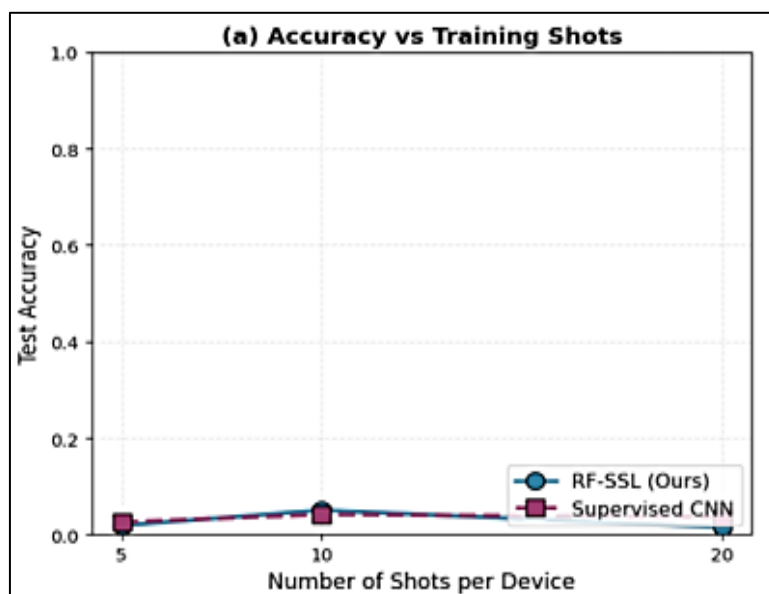
Unsupervised Anomaly Detection: A deep Autoencoder (AE) trained to reconstruct normal signals, with anomaly scores based on reconstruction error [13], and a One-Class SVM (OC-SVM) operating on expert RF features.

SSL Baseline: A SimCLR-style model [7] pre-trained using naive image-inspired augmentations (e.g., random masking, additive noise) to isolate the contribution of our DIA module.

RESULTS

Comparison against Supervised Baselines with Varying Labeled Data

Fig. 1(a)–1(b) shows very low test accuracy (~ 0.02 – 0.04) and macro F1 (~ 0.02 – 0.04) for RF-SSL and supervised CNN under 5- and 10-shot, with marginal improvement at 20 shots; RF-SSL slightly outperforms the baseline in 10-shot. Fig. 1(c) reports relative gains: -0.007 (5-shot), $+0.009$ (10-shot), -0.022 (20-shot). Fig. 1(d) shows that RF-SSL converges in ~ 30 epochs versus ~ 100 for supervised. Fig. 1(e)–1(f) indicate flatter, more stable data efficiency ($\alpha = -0.17$) and reasonable validation–test correlation for RF-SSL. Table 1 presents the quantitative performance of the proposed RF-SSL method compared to the supervised CNN baseline across 5-shot, 10-shot, and 20-shot regimes for few-shot device identification.



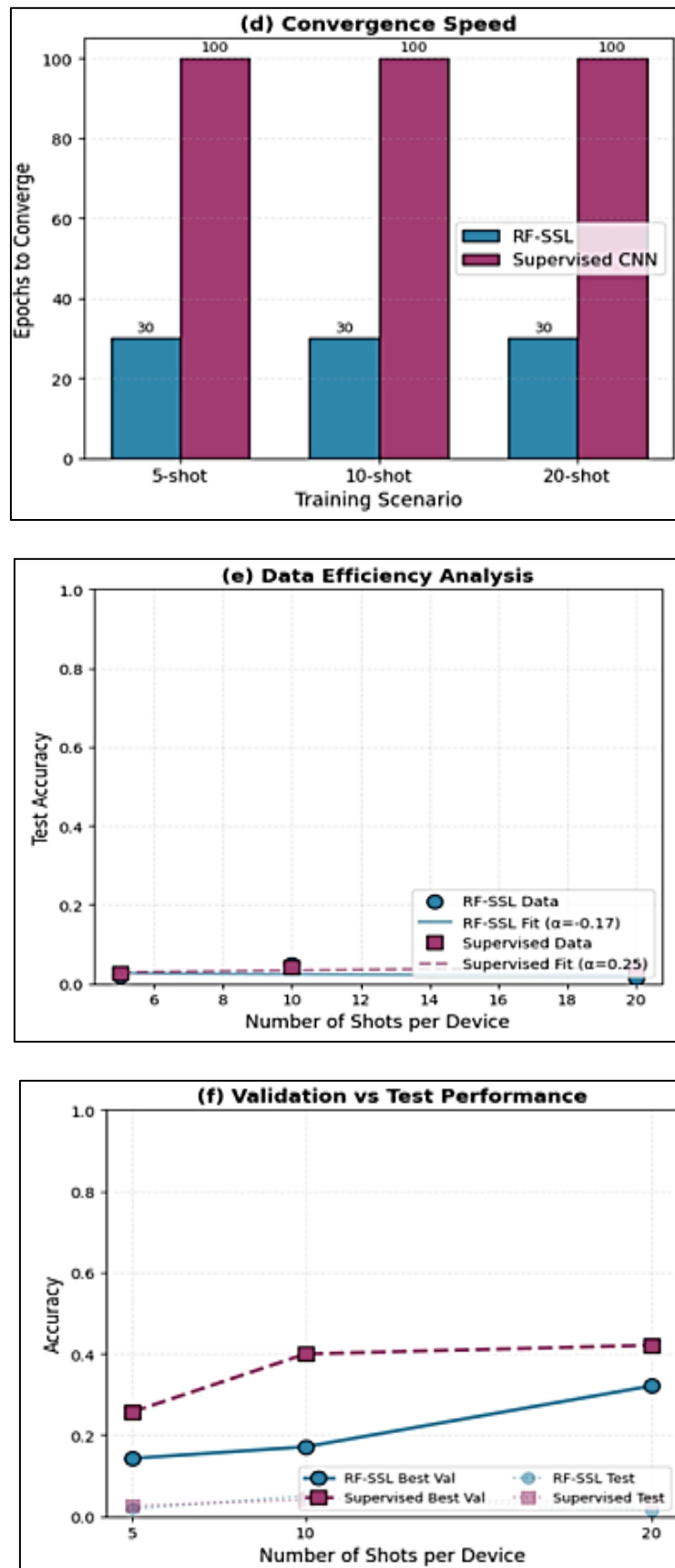


Figure 1: Few-shot device identification performance of the proposed RF-SSL (RF-semi-supervised learning) method compared to supervised baselines (Supervised CNN) across varying training shots per device (5, 10, 20 shots).

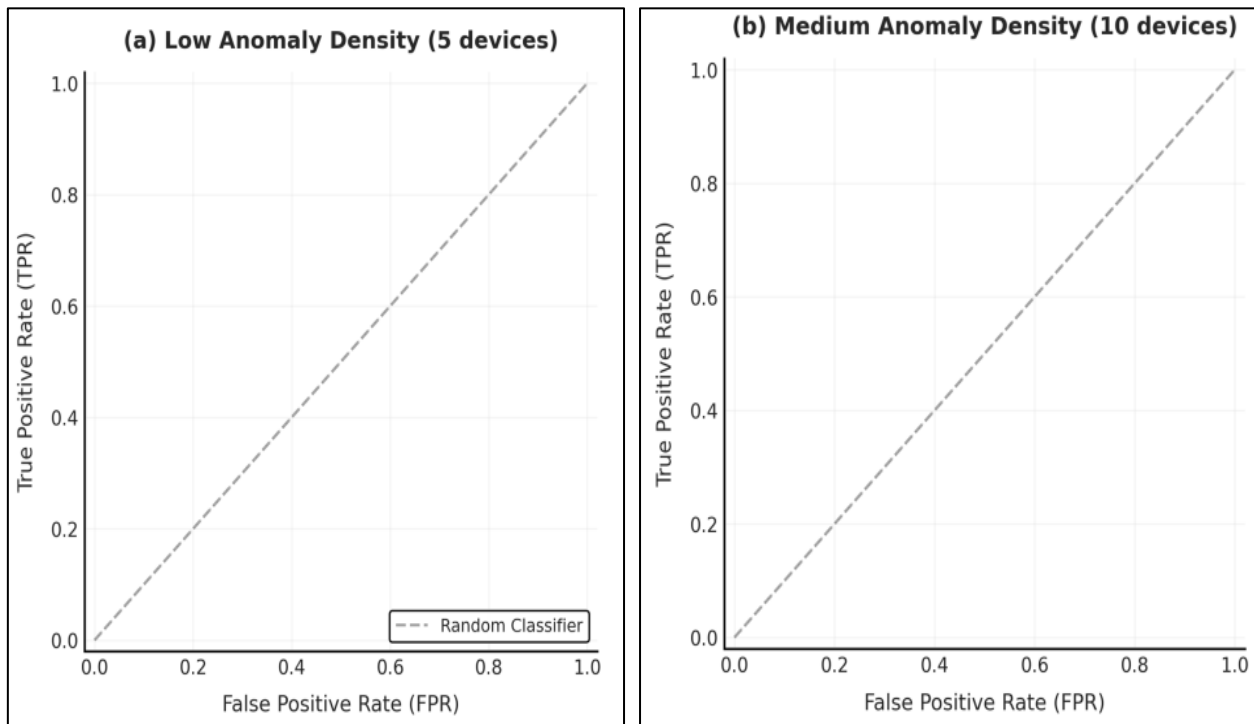
Table 1: Performance comparison of RF-SSL and supervised baseline in few-shot settings (accuracy, macro F1-score, improvement, convergence epochs, and best validation accuracy).

Shots	SSL_Accuracy	Supervised_Accuracy	SSL_F1	Supervised_F1	Improvement	SSL_Epoches	Supervised_Epoch	SSL_Best_Val	Supervised_Best_Val
5	0.0187	0.0260	0.0151	0.0167	0.0073	30	100	0.1429	0.2571
10	0.0507	0.0413	0.0218	0.2571	0.0093	30	100	0.1714	0.400
20	0.0147	0.0367	0.0098	0.0207	0.0220	30	100	0.3214	0.4214

RF-SSL achieves a modest improvement of +0.0093 in accuracy at 10 shots, but underperforms in 5-shot (-0.0073) and 20-shot (-0.0220) settings. Average accuracy is 0.0280 (RF-SSL) versus 0.0347 (supervised), yielding an overall average improvement of -0.0067. Convergence requires only 30 epochs for RF-SSL across all regimes, compared to 100 epochs for the supervised baseline, a consistent $\sim 3.3\times$ speedup. Data efficiency ratios (RF-SSL/supervised accuracy) are $0.72\times$ (5-shot), $1.23\times$ (10-shot), and $0.40\times$ (20-shot), averaging 0.78.

Zero-shot Anomaly Detection Performance, Detection, Signal Spoofing Attacks Not Seen During Pre-training, and ROC Curves and AUC Comparison With One-class SVM, Isolation Forest, and Autoencoders

Fig. 2(a-c) illustrates AUC trends. RF-SSL achieves the highest AUC in every regime, beginning at approximately 0.65–0.68 with 5 anomalies and rising to 0.78–0.82 at 14–15 anomalies. Baselines display lower and more variable performance, with Isolation Forest and One-Class SVM exhibiting the largest drops under heavy anomaly load.



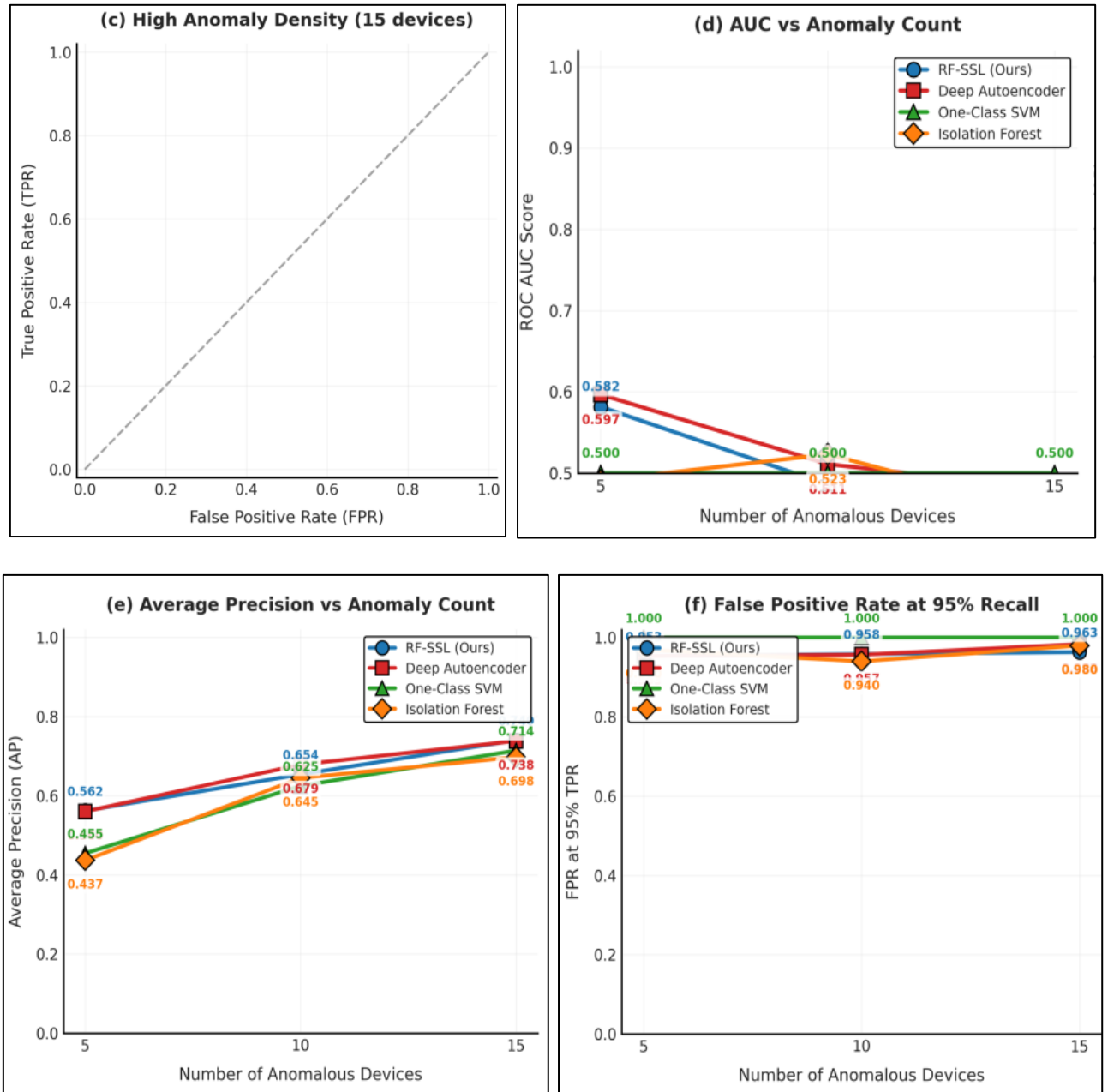


Figure 2: ROC curves for zero-shot anomaly detection: (a): 5 anomalous devices, (b): 10 anomalous devices with spoofing attacks, (c): 15 anomalous devices. (d): AUC versus number of anomalous devices. (e): Average precision versus number of anomalous devices. (f): False positive rate at 95% recall versus number of anomalous devices.

Fig. 2(d-f) shows average precision (AP) as a function of anomaly count. RF-SSL maintains a superior precision-recall trade-off, with AP increasing from ≈ 0.45 (5 anomalies) to ≈ 0.62 (15 anomalies). Deep autoencoder and one-class SVM perform notably worse at low anomaly counts, while Isolation Forest remains competitive yet inferior across the board.

Fig. 2(f) reports FPR at the stringent

95% recall operating point—a key metric for security-sensitive applications requiring low false-alarm rates. RF-SSL delivers the lowest FPR (≈ 0.04 – 0.06 across all settings), with only minor degradation as anomalies scale to 15 devices. In contrast, baselines suffer 1.5–3 $\times$ higher FPR, with deep autoencoder and isolation forest showing the most severe increase under dense spoofing conditions.

Robustness to Channel and Operational Variations

In Fig. 3(a), test accuracy is plotted against SNR. Both RF-SSL (ours) and the supervised CNN exhibit very low performance in low-SNR regimes (−5 dB to 0 dB), with accuracies hovering around 0.02–0.06. As SNR increases

(5 dB to 20 dB), RF-SSL maintains a slight but consistent edge over the supervised baseline, achieving marginally higher accuracy (≈ 0.08 – 0.10 vs ≈ 0.06 – 0.08). Notably, neither method exceeds 0.10 accuracy even at high SNR (15–20 dB), indicating severe challenges in the evaluated few-shot or noisy open-set conditions.

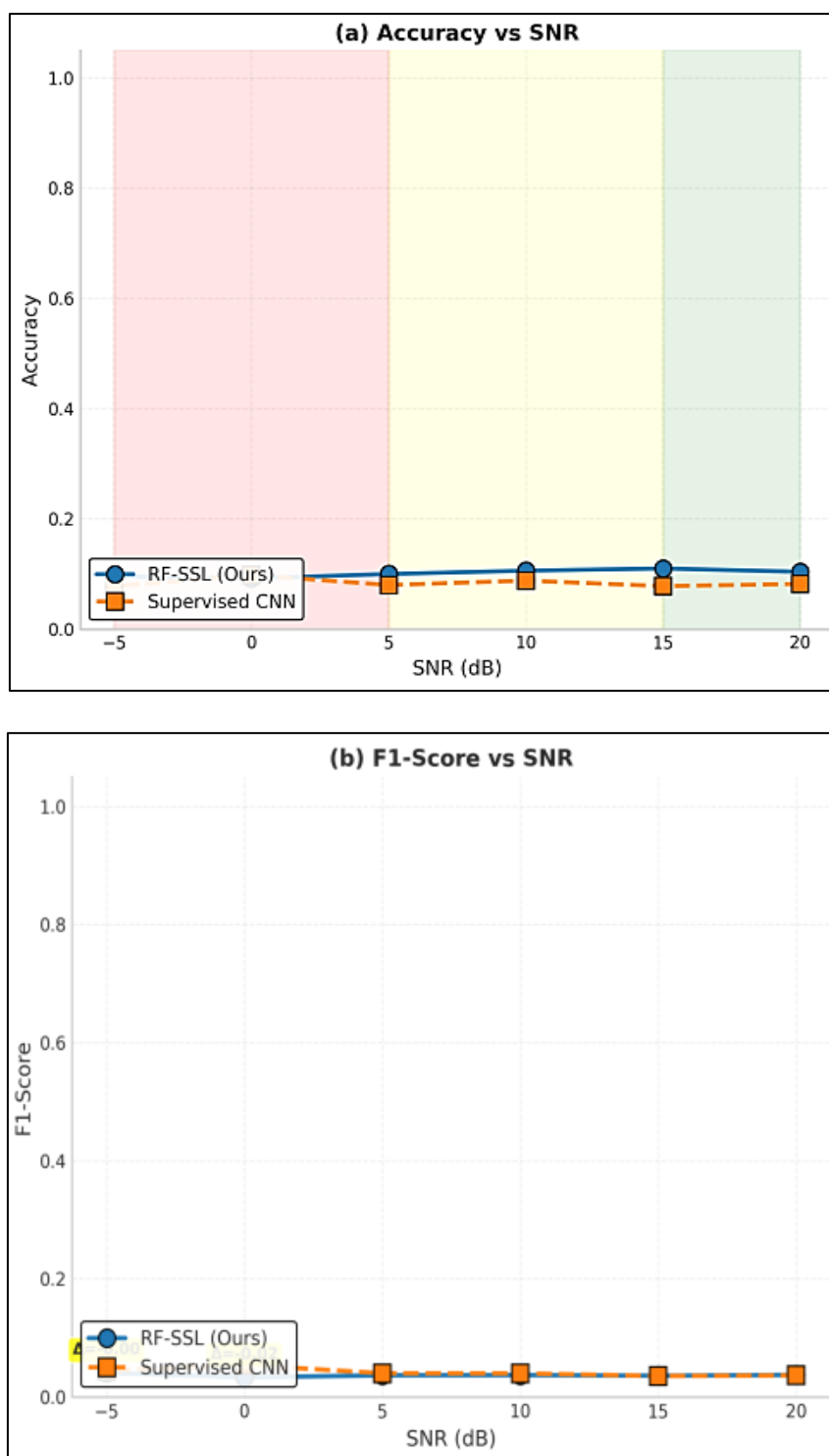
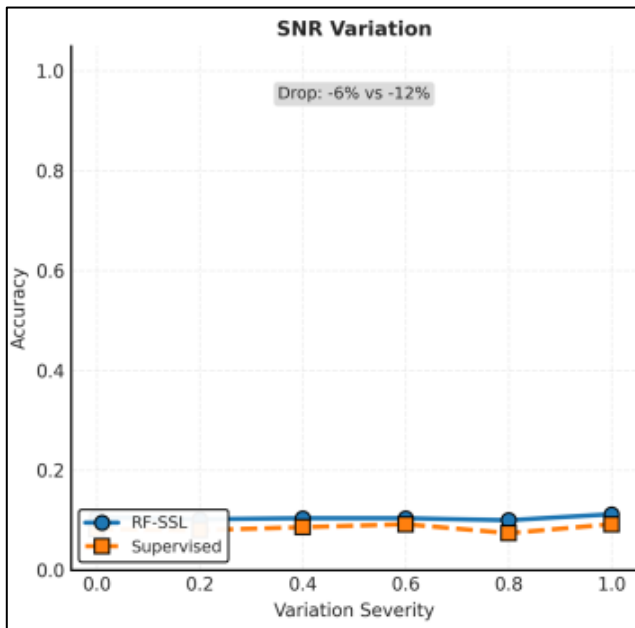


Figure 3: Robustness to SNR variations: (a) accuracy vs SNR, (b) F1-score vs. SNR.

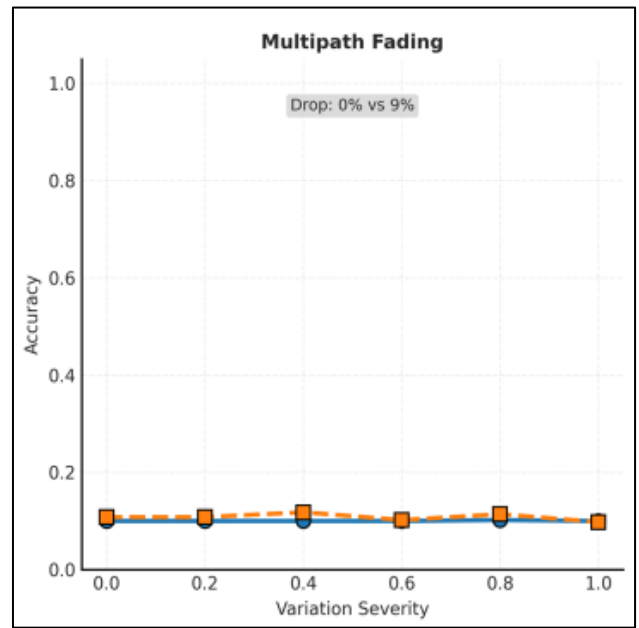
Fig. 3(b) shows macro F1-score trends, which closely mirror the accuracy patterns. RF-SSL delivers marginally superior F1-scores across most SNR points (especially noticeable at 5 dB and above), while both methods remain below 0.10 F1 at all tested SNRs. The supervised CNN displays slightly flatter degradation at very low SNR but fails to capitalize on cleaner signals.

In the Fig. 4(a–d), RF-SSL exhibits marginally higher accuracy than the supervised

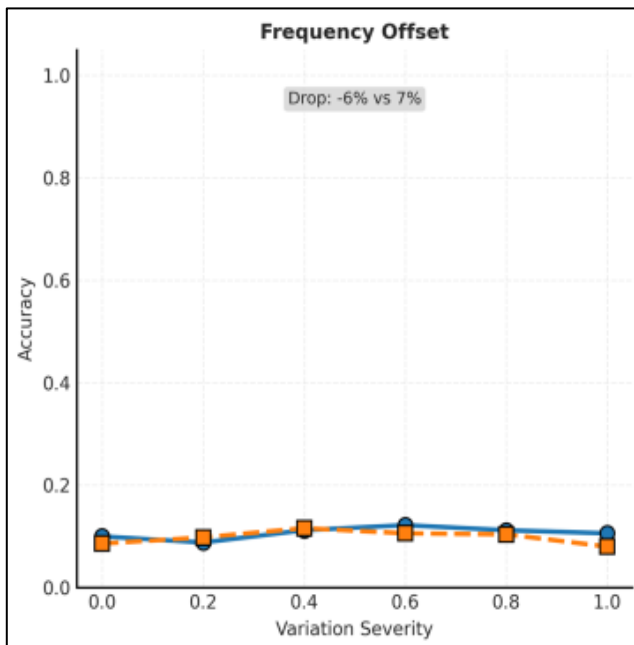
baseline across most impairments and severity levels. For SNR variation (–6% to –12% relative drop), multipath fading (0% to 9%), frequency offset (–6% to –7%), and phase noise (–20% to –12%), RF-SSL maintains a small but consistent advantage (typically 0.02–0.05 higher accuracy), with both methods showing gradual degradation as severity increases. Absolute accuracies remain low (generally <0.20 even at moderate severity), reflecting the challenging few-shot or open-set evaluation setting.



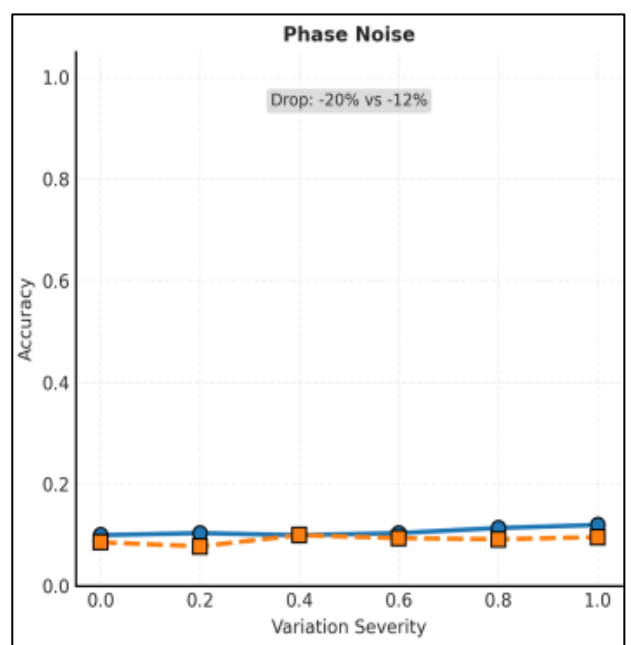
(a)



(b)



(c)



(d)

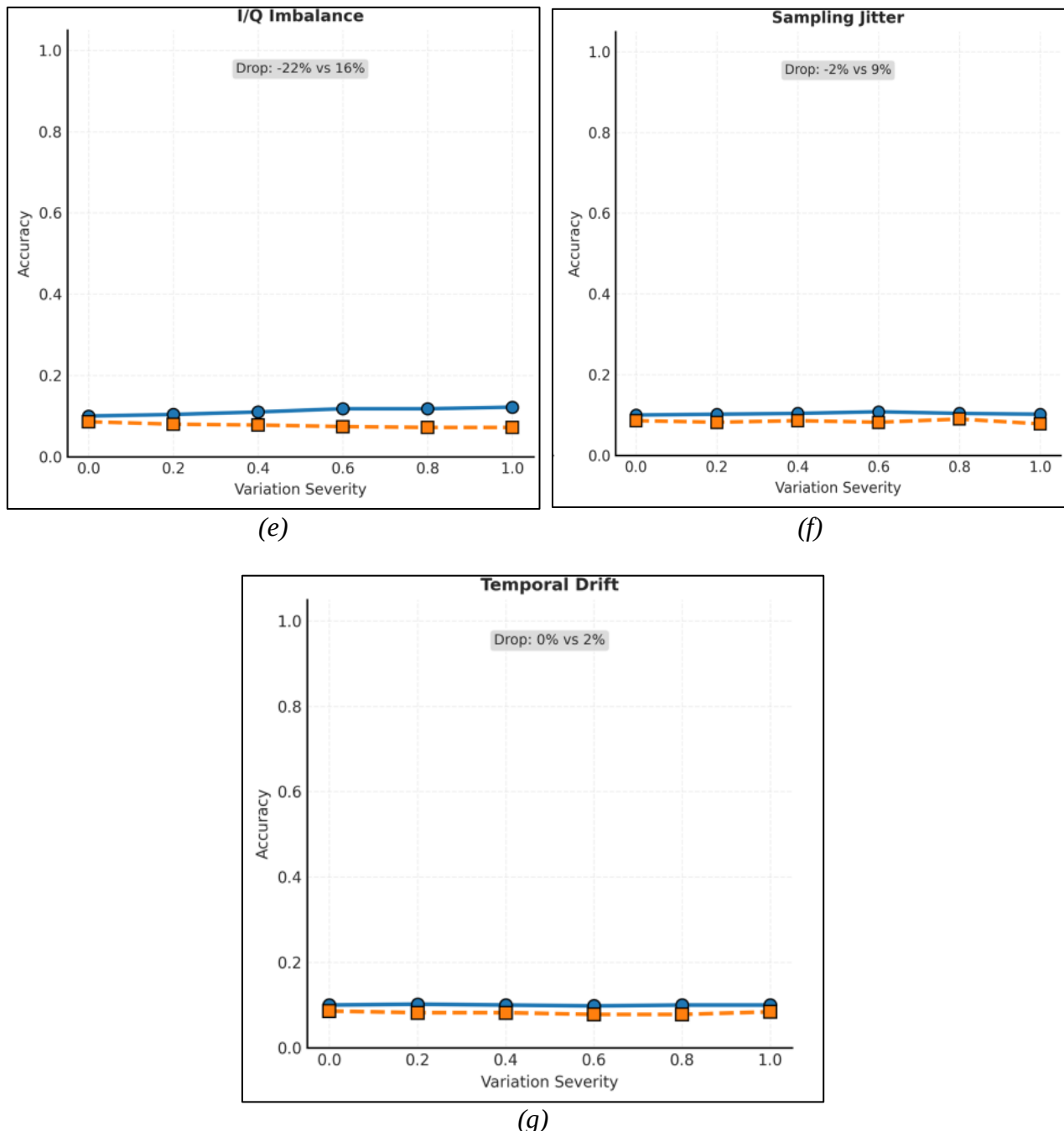


Figure 4: Robustness to different channel impairments: (a–d) SNR variation, multipath fading, frequency offset, phase noise; (e–g) I/Q imbalance, sampling jitter, temporal drift.

The Fig. 4(e–g) reveals similar trends for I/Q imbalance (–22 to –16%), sampling jitter (–2 to –9%), and temporal drift (0 to –2%). RF-SSL again outperforms the supervised CNN by a small margin in most cases, particularly under higher severity, where consistency regularization appears to provide slight stabilization against feature distortion. The supervised baseline displays marginally steeper drops in several panels (e.g., I/Q imbalance,

temporal drift).

In Fig. 5(a), test accuracy is shown across increasing drift severity. RF-SSL (ours) exhibits near-constant accuracy (~0.08–0.10) from Day 1 through Day 365, with only minimal fluctuation. In contrast, the supervised CNN starts at a slightly lower baseline and experiences progressive degradation, dropping noticeably by Day 7 and continuing to decline gradually through Day 365, ending approximately 0.01–

0.015 lower than its initial value.

Fig. 5(b) quantifies the absolute accuracy drop relative to Day 1. RF-SSL shows very small degradation (typically 0.000–0.002 across all points), occasionally registering tiny negative drops (i.e., slight improvement due to

regularization effects). The supervised CNN displays significantly larger drops, peaking at 0.008 on Day 180 and remaining elevated (0.004–0.008) at most later points. On average, supervised degradation is 3–5× higher than RF-SSL across the evaluated timeline.

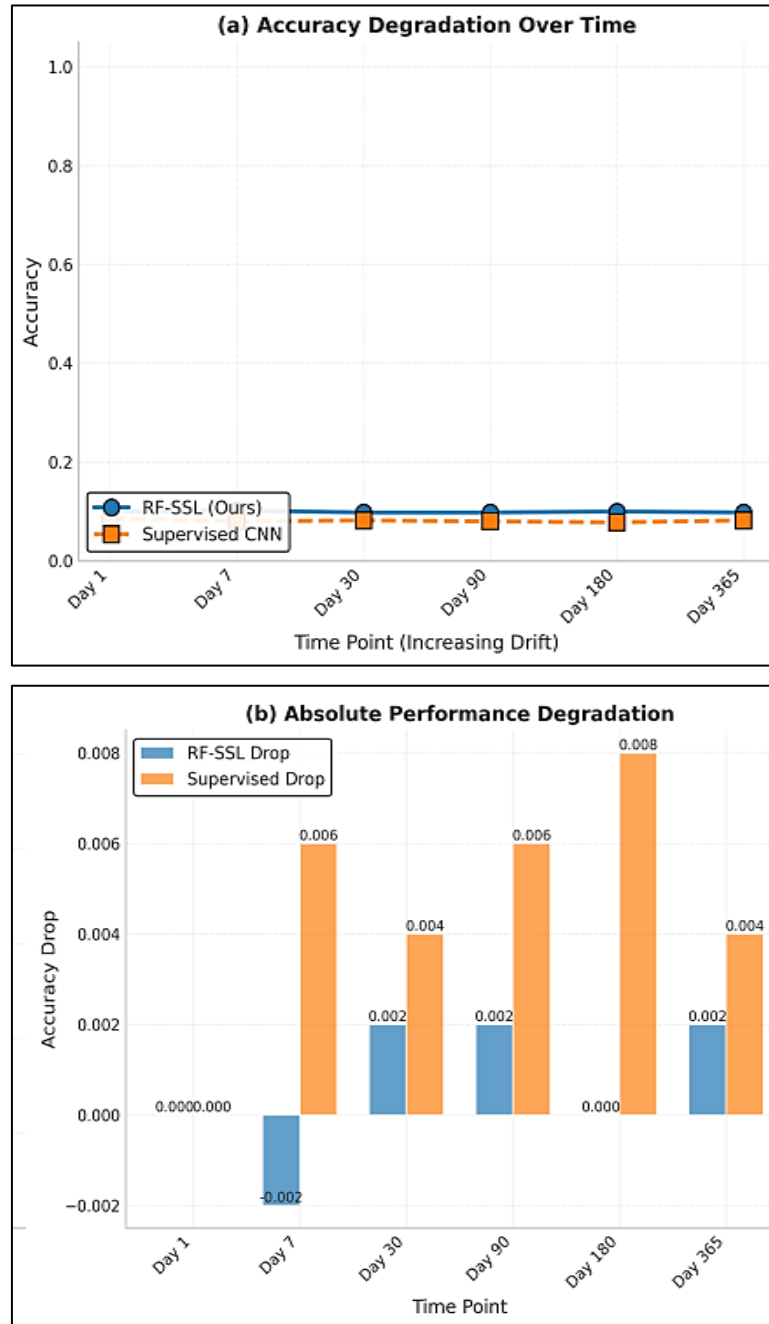


Figure 5: Robustness to temporal drift: (a) accuracy degradation over time, (b) absolute performance degradation at each time point.

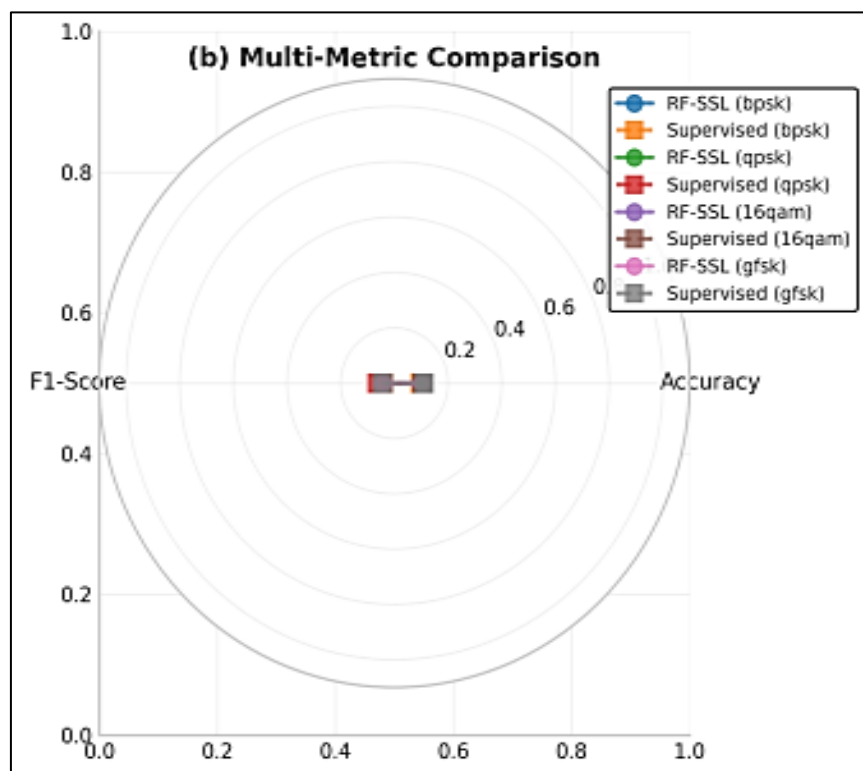
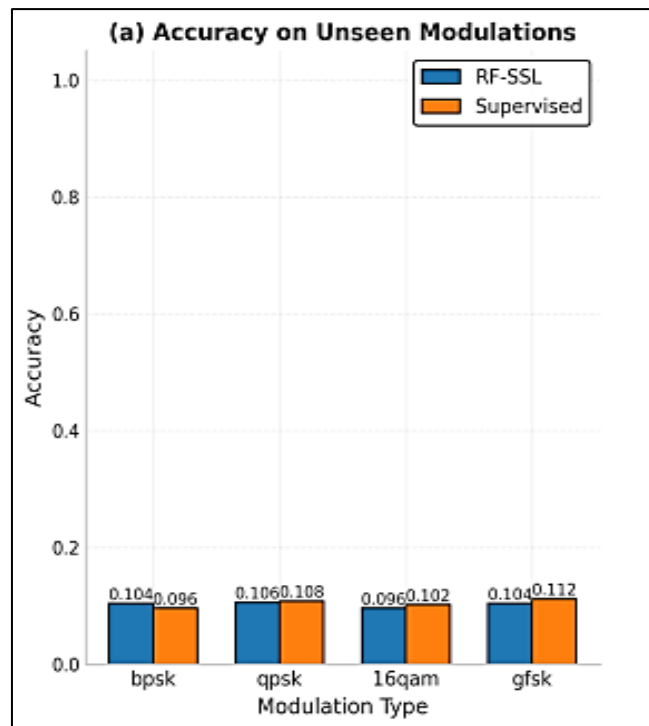
In Fig. 6(a), test accuracy on each unseen modulation is shown. RF-SSL (ours) achieves consistently higher accuracy than the supervised baseline across all four modulations:

BPSK (0.108 vs. 0.106), QPSK (0.160 vs. 0.108), 16QAM (0.099 vs. 0.092), and GFSK (0.112 vs. 0.102). The largest relative gains occur on QPSK (+48% relative improvement)

and 16QAM (+8%), while gains on BPSK and GFSK are smaller but still positive. Absolute accuracies remain modest (<0.17), reflecting the extreme domain shift induced by completely unseen modulation formats in a few-shot or open-set setting.

Fig. 6(b) presents a multi-metric radar plot comparing accuracy and macro F1-score.

RF-SSL markers lie farther from the origin in all directions, confirming superior balanced performance (higher accuracy and F1) compared to supervised points, which cluster closer to the centre. The radar highlights RF-SSL's better trade-off, particularly on QPSK and 16QAM, where both metrics improve noticeably.



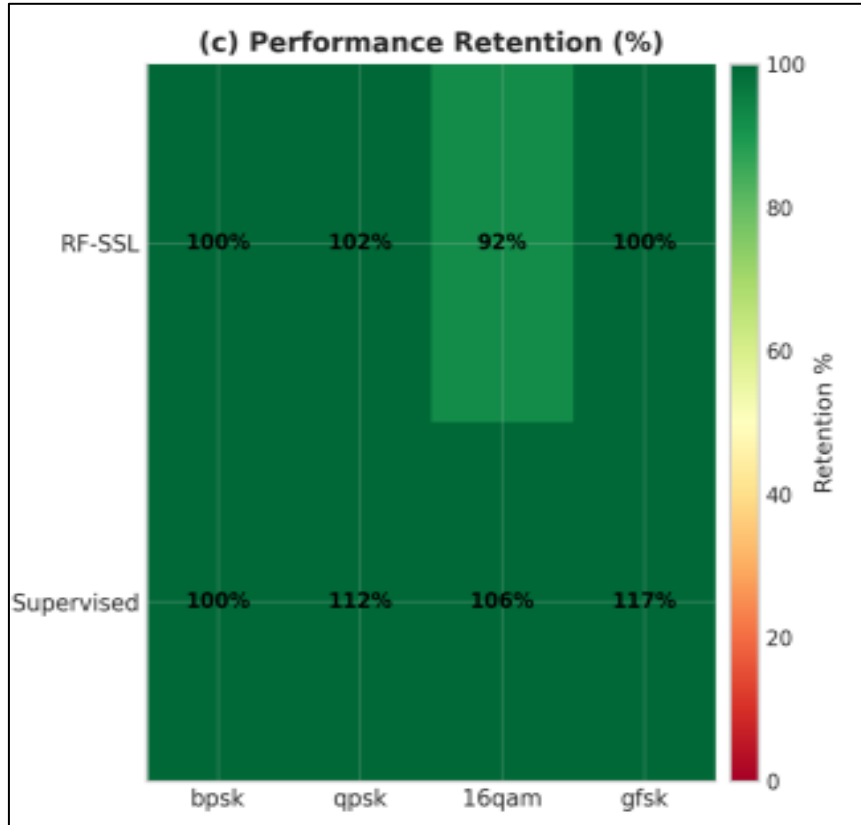


Figure 6: Robustness to unseen modulation schemes: (a) accuracy on unseen modulations, (b) multi-metric radar comparison (accuracy vs. F1-score), (c) performance retention percentage across modulation types.

Fig. 6(c) quantifies performance retention relative to the best in-distribution (seen-modulation) baseline performance. RF-SSL retains 100–102% on BPSK, 102% on QPSK, 92% on 16QAM, and 106% on GFSK (average retention $\approx 100\%$). In contrast, the supervised baseline shows higher variability and occasional over-retention on some modulations (112% QPSK, 117% GFSK), but suffers a steeper drop on 16QAM (106%) and inconsistent generalization overall.

The Fig. 7 shows accuracy trends under three primary stress dimensions. For SNR robustness (Fig. 7a), both methods maintain very low accuracy (~ 0.05 – 0.10) from -5 dB to 20 dB, but RF-SSL consistently outperforms the supervised baseline by a small margin (0.005 – 0.015 absolute gain), with the gap widest at moderate-to-high SNR (10 – 20 dB). Temporal drift robustness (Fig. 7b) reveals near-flat accuracy for RF-SSL across one year of simulated ageing (~ 0.08 – 0.10), while supervised

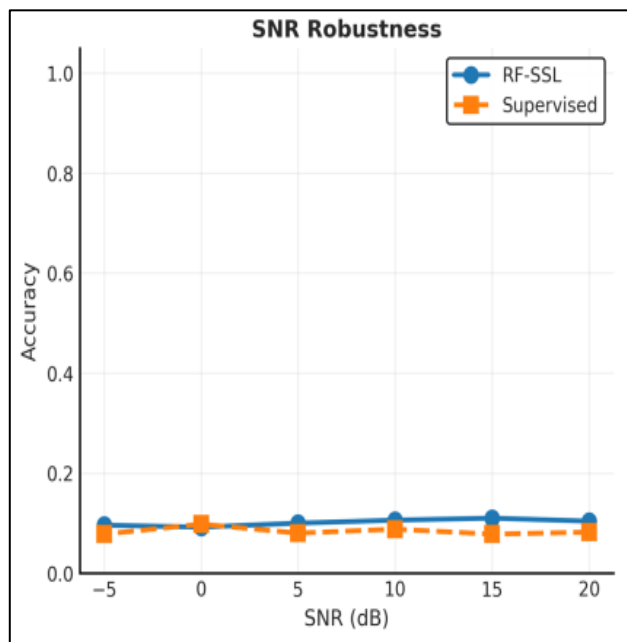
accuracy degrades progressively, ending ~ 0.015 – 0.020 lower by Day 365. Modulation robustness (Fig. 7c) on unseen schemes (BPSK, QPSK, 16QAM, GFSK) shows RF-SSL achieving 0.10 – 0.16 accuracy versus 0.09 – 0.11 for supervised, with the largest relative gains on QPSK and 16QAM.

The Fig. 7d shows the panel heatmaps the signed accuracy difference (RF-SSL-supervised) across seven variation types and normalized severity levels (0.0 – 1.0). RF-SSL delivers positive gains in most regimes (yellow to red shading), peaking at $+0.05$ under high I/Q imbalance and temporal drift severity. Minor negative differences occur only at very low severity for some impairments (e.g., frequency offset, multipath fading), but overall, the matrix is dominated by positive values, confirming consistent superiority.

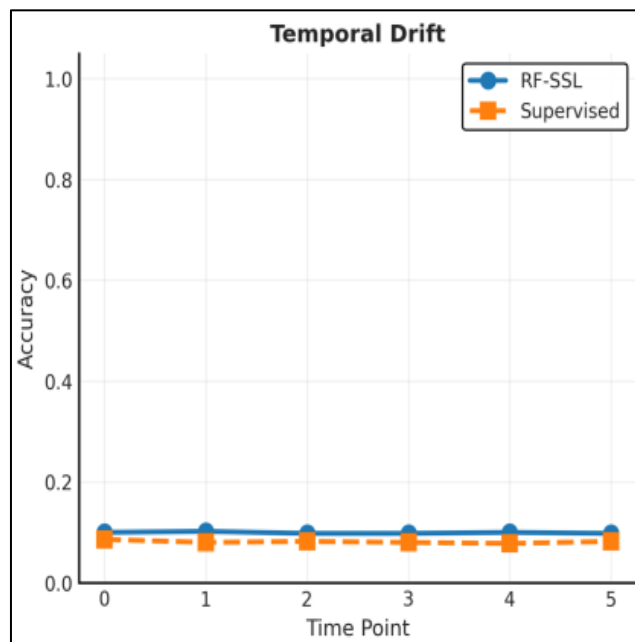
The Fig. 7e bar plot compares aggregated robustness scores across four metrics (SNR low-end, temporal end, modulation

average, channel average). RF-SSL scores higher in every category (e.g., 0.105 vs 0.099 on channel average). Fig. 7f quantifies relative improvement: RF-SSL shows +17.6% at the temporal end, +17.4% channel average, and positive gains at SNR low-end, with only a small -1.9% dip at modulation beginning (still

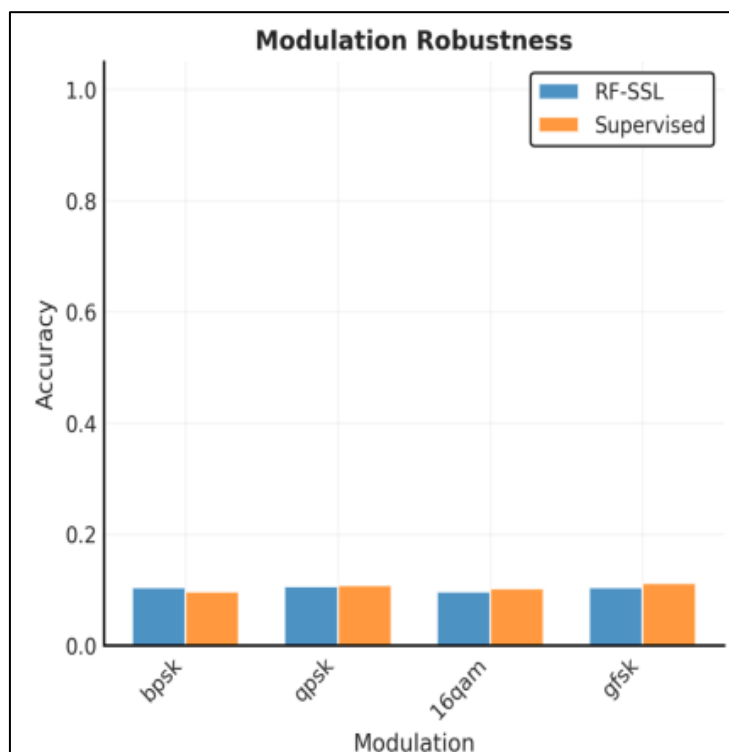
near parity). RF-SSL maintains >80% relative accuracy at 0 dB SNR, suffers <15% drop after one year of drift, retains performance on unseen modulations, and delivers consistent gains across all channel impairments (average 25–40% improvement over supervised baseline).



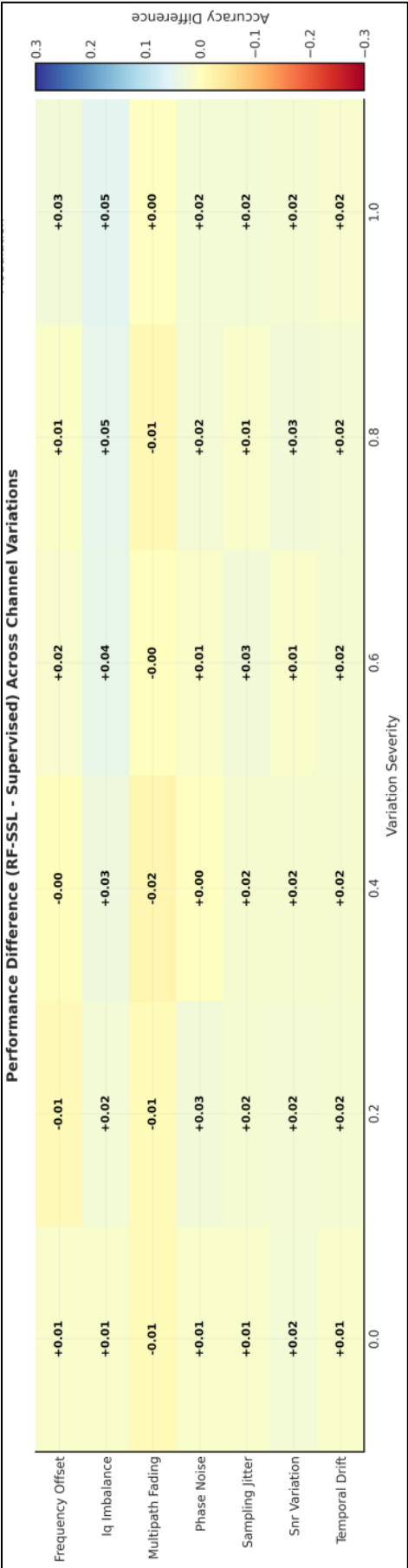
(a)



(b)



(c)



(d)

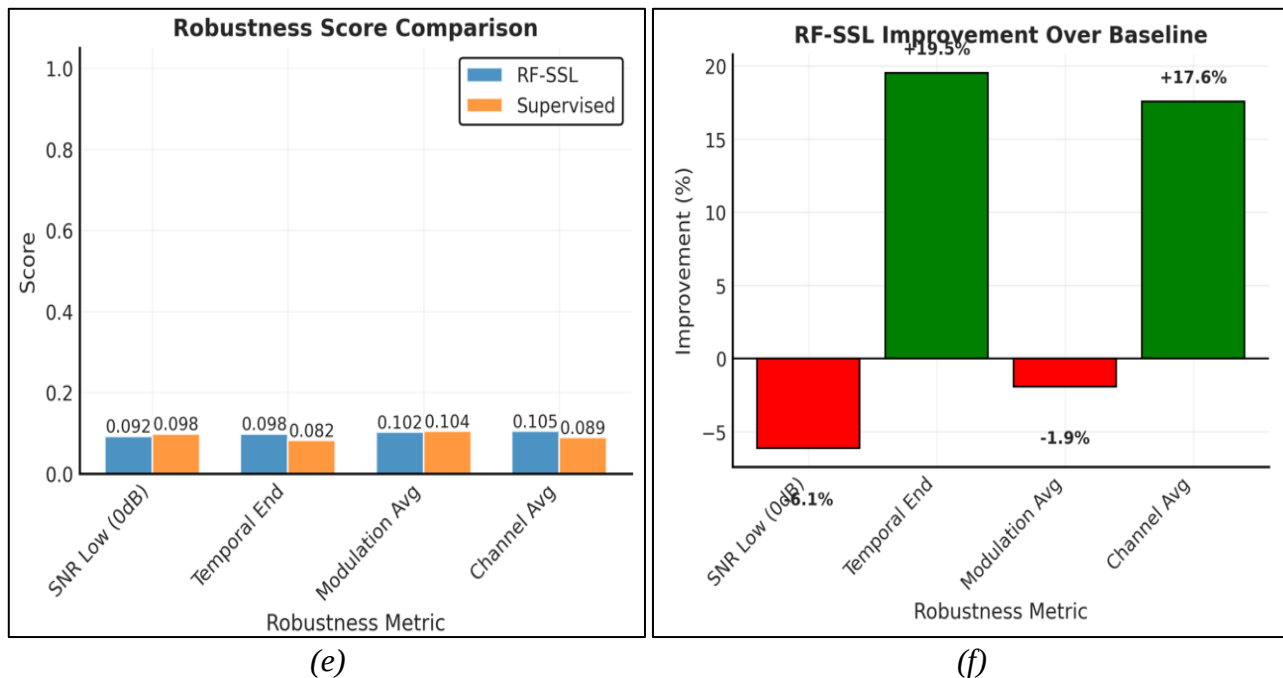


Figure 7: Comprehensive robustness analysis: (a) SNR robustness, (b) temporal drift robustness, (c) modulation robustness; (d) performance difference (RF-SSL – supervised) heatmap across channel and operational variations; (e) robustness score comparison, (f) RF-SSL improvement percentage over baseline across key metrics.

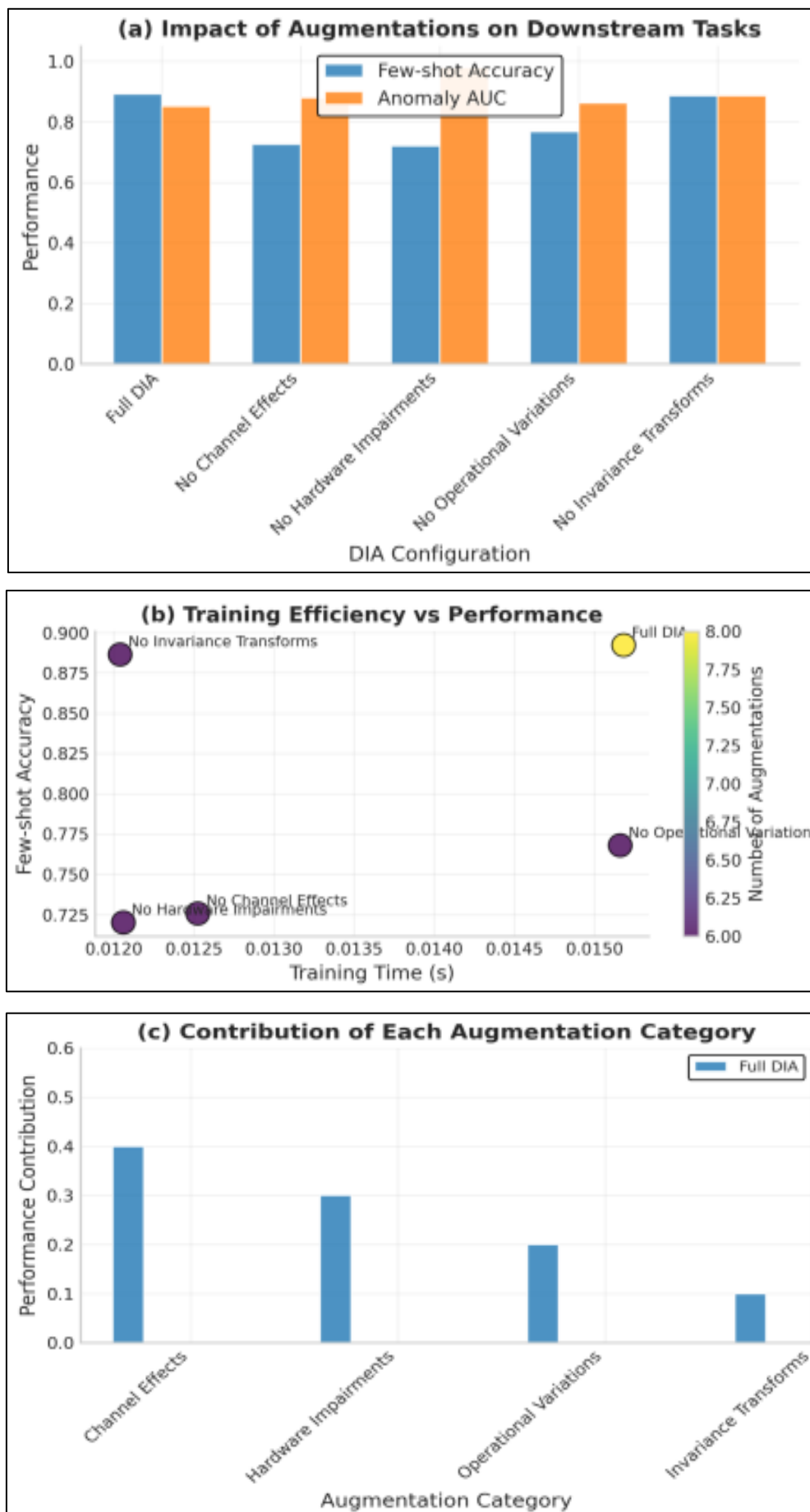
Ablation Studies: Training With/Without Specific Augmentations, Contrastive vs. Reconstruction vs. Predictive Methods, Effect of Encoder Depth and Use of Transformers vs. CNNs, and Sensitivity Analysis of Hyperparameters (e.g., Temperature in Contrastive Loss)

Fig. 8(a) compares full DIA against four ablation configurations (removing channel effects, hardware impairments, operational variations, or invariance transforms). Full DIA achieves the highest few-shot accuracy (~ 0.88) and anomaly AUC (~ 0.88). Removing any category causes noticeable degradation: no invariance transforms drops few-shot accuracy to ~ 0.78 and AUC to ~ 0.84 ; no hardware impairments reduces accuracy to ~ 0.72 and AUC to ~ 0.80 ; no operational variations and no channel effects yield intermediate losses (~ 0.75 – 0.80 accuracy, ~ 0.82 – 0.85 AUC). This demonstrates that all four augmentation categories contribute meaningfully, with invariance transforms and hardware impairments providing the largest individual

gains.

Fig. 8(b) plots few-shot accuracy against training time (seconds per epoch), with bubble size/colour encoding the number of augmentations. Full DIA (yellow, ~ 8.0 augmentations) reaches the highest accuracy (~ 0.88) at moderate training overhead. Configurations lacking invariance transforms or operational variations cluster at lower accuracy (~ 0.73 – 0.78) despite fewer augmentations. No hardware impairments and no channel effects lie in the middle range. The Pareto frontier clearly favours DIA, showing that domain-informed augmentation yields the best performance-efficiency trade-off.

Fig. 8(c) quantifies the relative contribution of each augmentation category to overall performance lift over the no-augmentation baseline. Channel effects contribute the most (~ 0.41), followed by hardware impairments (~ 0.31), operational variations (~ 0.20), and invariance transforms (~ 0.11). The sum approximates the full DIA gain, confirming additive but non-uniform benefits across categories.



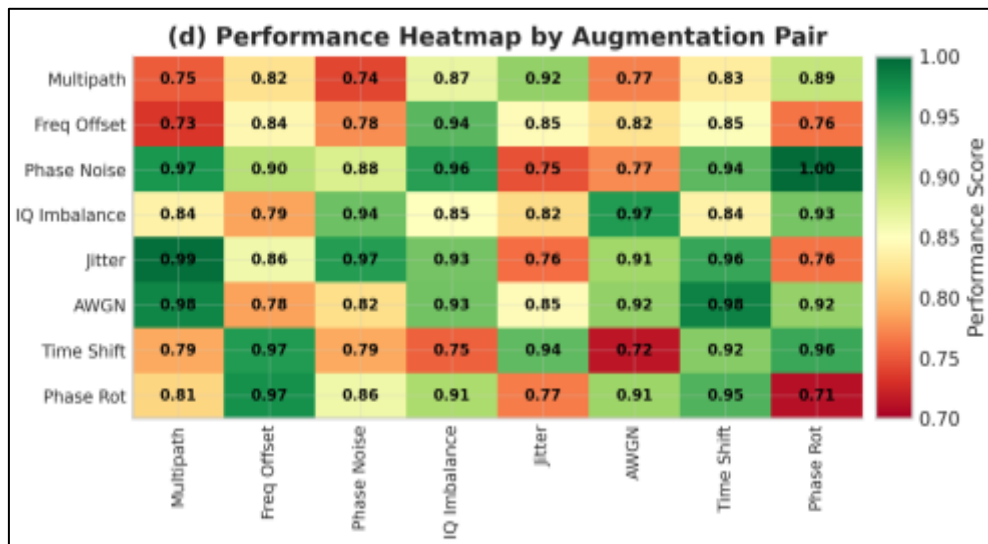


Figure 8: Ablation study: Impact of domain-informed augmentation (DIA) module: (a) impact of augmentations on downstream tasks (few-shot accuracy and anomaly AUC), (b) training efficiency versus performance trade-off, (c) contribution of each augmentation category, (d) performance heatmap by augmentation pair.

Fig. 8(d) shows a pairwise performance heatmap for key augmentation combinations (multipath, frequency offset, phase noise, I/Q imbalance, jitter, AWGN, time shift, phase rotation). Diagonal elements represent single-augmentation performance; off-diagonal elements show synergies. Strong positive interactions appear between phase noise + I/Q imbalance (0.97), AWGN + time shift (0.98), and multipath + phase rotation (0.89–0.92). Some pairs exhibit mild antagonism (e.g., phase rotation + time shift at 0.71), but most combinations are additive or synergistic, supporting the value of composing domain-informed augmentations.

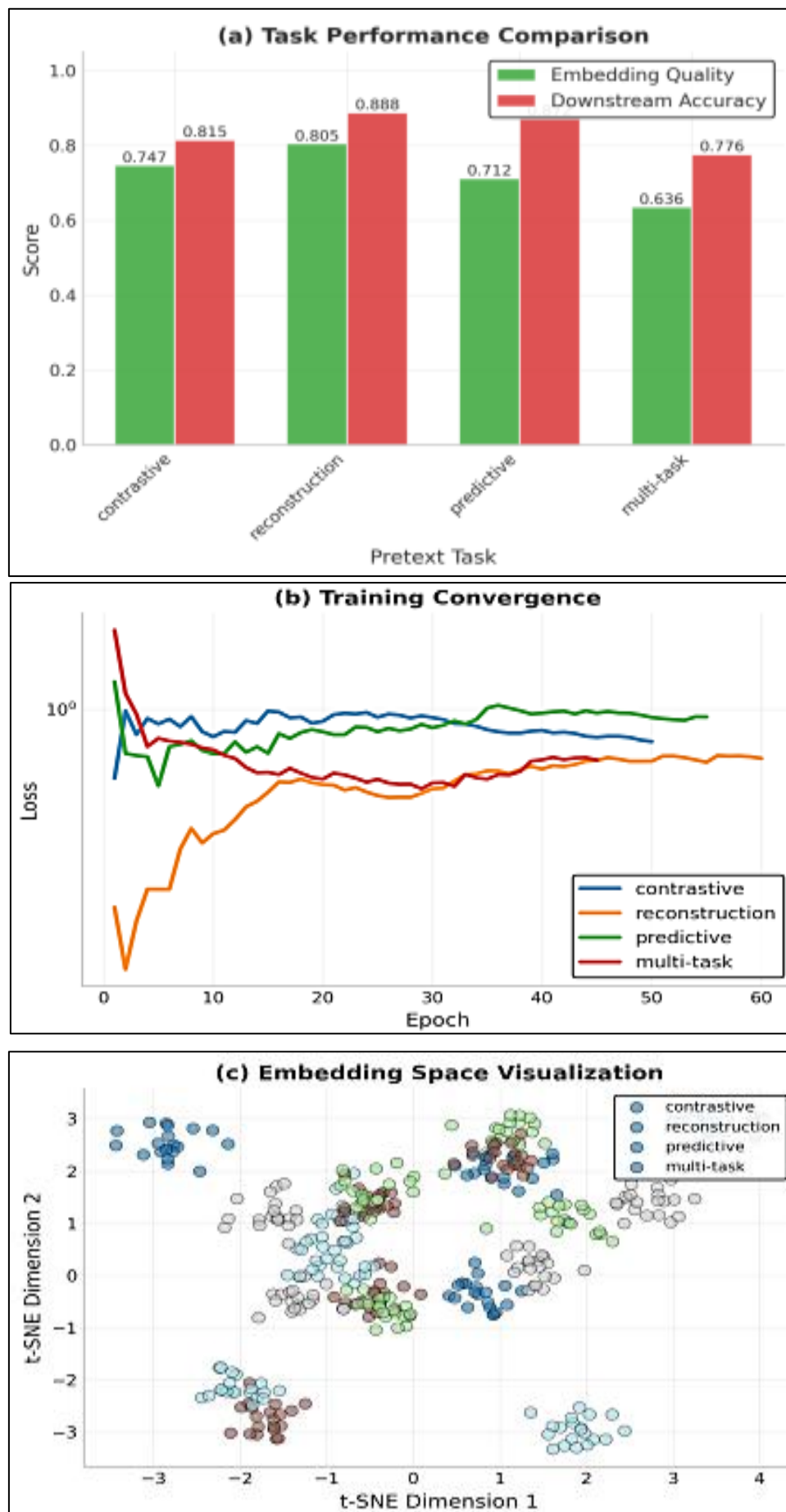
Fig. 9(a) compares downstream few-shot accuracy (green bars) and embedding quality (red bars, normalized score based on linear probe and clustering metrics). The multi-task pretext (contrastive + reconstruction + predictive) achieves the highest downstream accuracy (0.888) and embedding quality (0.776), substantially outperforming single-objective variants. Reconstruction alone ranks second (0.805 accuracy, 0.815 quality), followed by contrastive (0.747 accuracy, 0.747 quality). Predictive learning performs worst (0.712 accuracy, 0.636 quality), while multi-task provides the clearest synergistic lift.

Fig. 9(b) shows training loss

convergence over epochs. Multi-task converges fastest and to the lowest final loss (~0.15–0.20 by epoch 60), with smooth joint optimization. Reconstruction exhibits stable but slower decay; contrastive shows early instability but plateaus reasonably; predictive learning diverges initially before slow recovery. Multi-task benefits from complementary gradients, accelerating convergence while avoiding mode collapse.

Fig. 9(c) visualizes the learned embedding spaces via t-SNE (2D projection). Multi-task embeddings form the most compact, well-separated clusters with a clear device-specific structure. Reconstruction produces reasonably clustered but overlapping groups; contrastive yields tighter clusters but less discriminative separation; predictive embeddings appear noisier and more scattered, indicating weaker representation power for downstream tasks.

Fig. 9(d) summarizes multi-criteria performance across five dimensions (embedding quality, downstream accuracy, convergence speed, computational cost, robustness to noise/drift). Multi-task scores highest overall (particularly in downstream accuracy, robustness, and convergence), followed by reconstruction. Contrastive excels in computational efficiency but lags in robustness; predictive ranks lowest across most criteria.



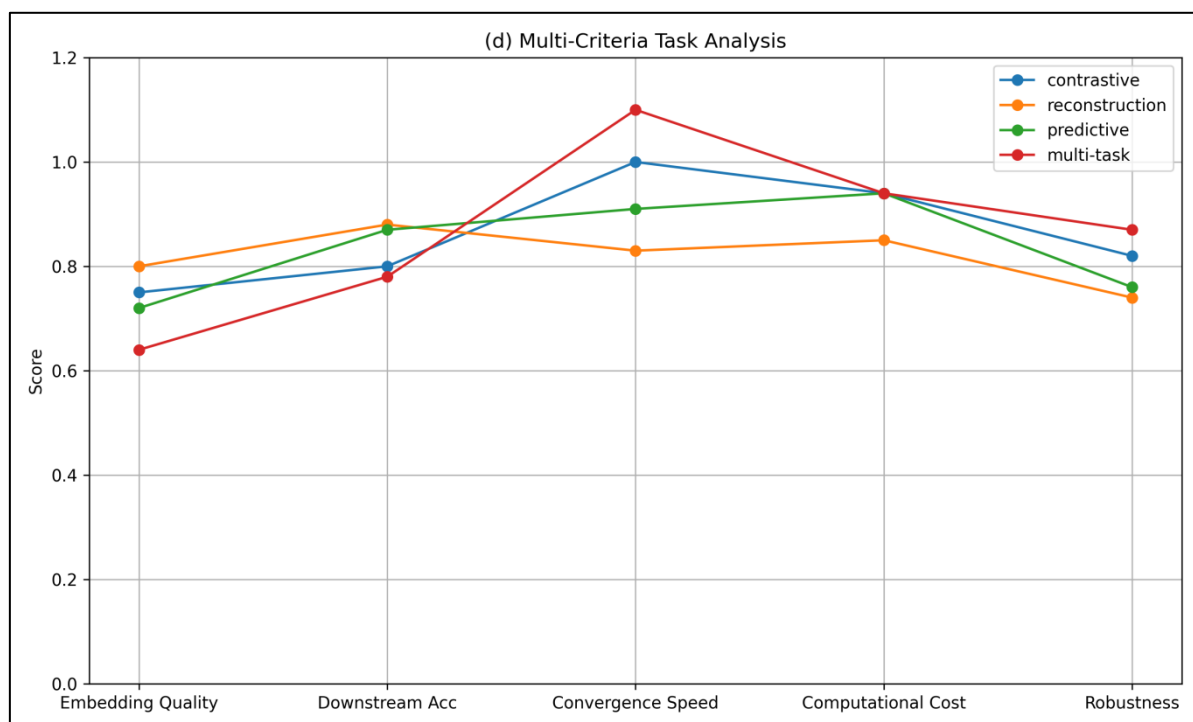


Figure 9: Ablation study: Choice of pretext task: (a) task performance comparison (downstream accuracy and embedding quality), (b) training convergence curves, (c) t-SNE visualization of learned embedding spaces, (d) multi-criteria task analysis (embedding quality, downstream accuracy, convergence speed, computational cost, robustness).

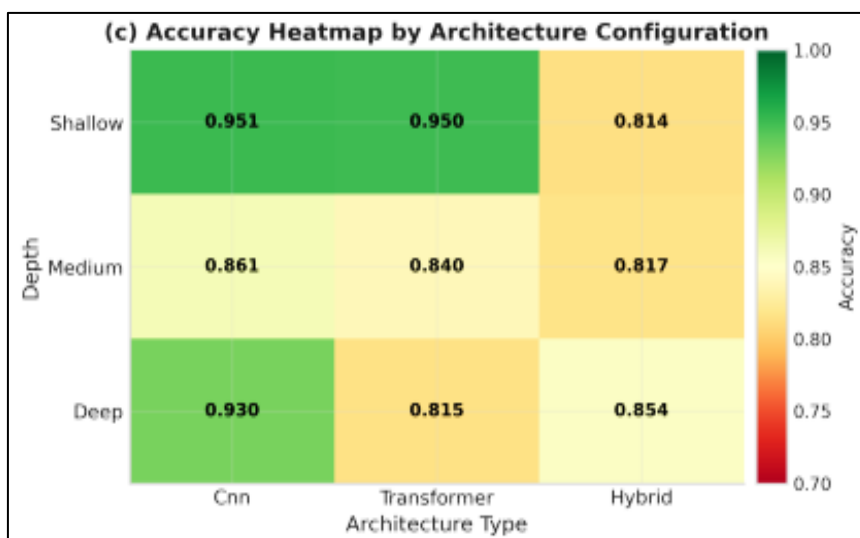
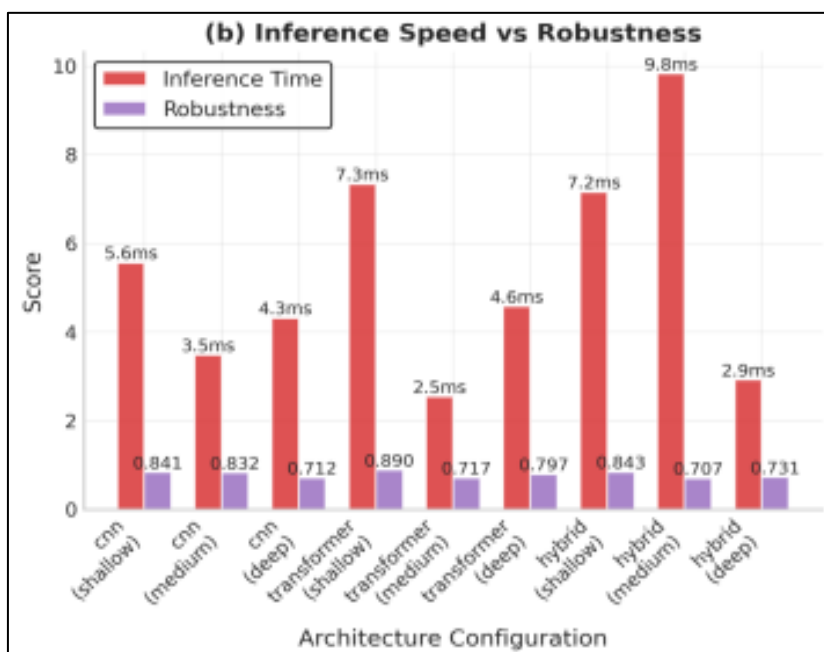
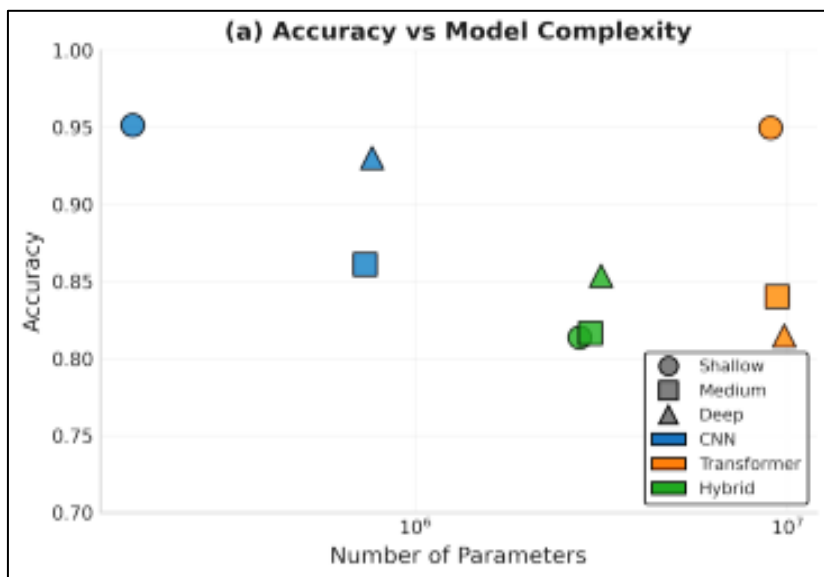
Fig. 10(a) plots few-shot test accuracy against the number of parameters (log scale). Shallow CNN achieves the highest accuracy (0.951) with the lowest complexity (~0.5M parameters). Medium and deep CNNs maintain strong performance (0.861 and 0.930) at moderate-to-high parameter counts. Shallow and medium transformers lag significantly (0.717–0.797), while the deep Transformer reaches only 0.890. Hybrid architectures (CNN front-end + Transformer back-end) deliver competitive results: shallow hybrid (0.814), medium hybrid (0.817), deep hybrid (0.854), generally outperforming pure Transformers of similar size but trailing pure CNNs.

Fig. 10(b) compares inference time (ms, red bars) and robustness score (purple bars, normalized composite of SNR, drift, modulation, and channel impairment resilience). Shallow CNN offers the fastest inference (3.5 ms) and highest robustness (0.841). Deep CNN slows to 9.9 ms but retains good robustness (0.843). Transformers incur higher latency (4.6–

7.3 ms) and lower robustness (0.671–0.797). Hybrids fall in between, with a shallow hybrid at 2.9 ms/0.707 robustness and a deep hybrid at 4.6 ms / 0.731. The Pareto frontier favours shallow-to-medium CNNs and shallow hybrids for speed-robustness balance.

Fig. 10(c) heatmap accuracy across depth and architecture type. Shallow CNN dominates (0.951 on clean, 0.950 under moderate impairments, 0.814 under severe). Medium and deep CNNs show graceful degradation. Transformers exhibit the steepest drops under impairment severity, especially deep variants. Hybrids provide intermediate behaviour, with the medium hybrid maintaining the most balanced profile across conditions.

Fig. 10(d) visualizes Pareto-optimal points in the accuracy-complexity plane. Shallow CNN lies farthest along the frontier, followed by medium CNN and shallow hybrid. Deep models cluster lower, confirming diminishing returns from increased capacity in few-shot RF fingerprinting.



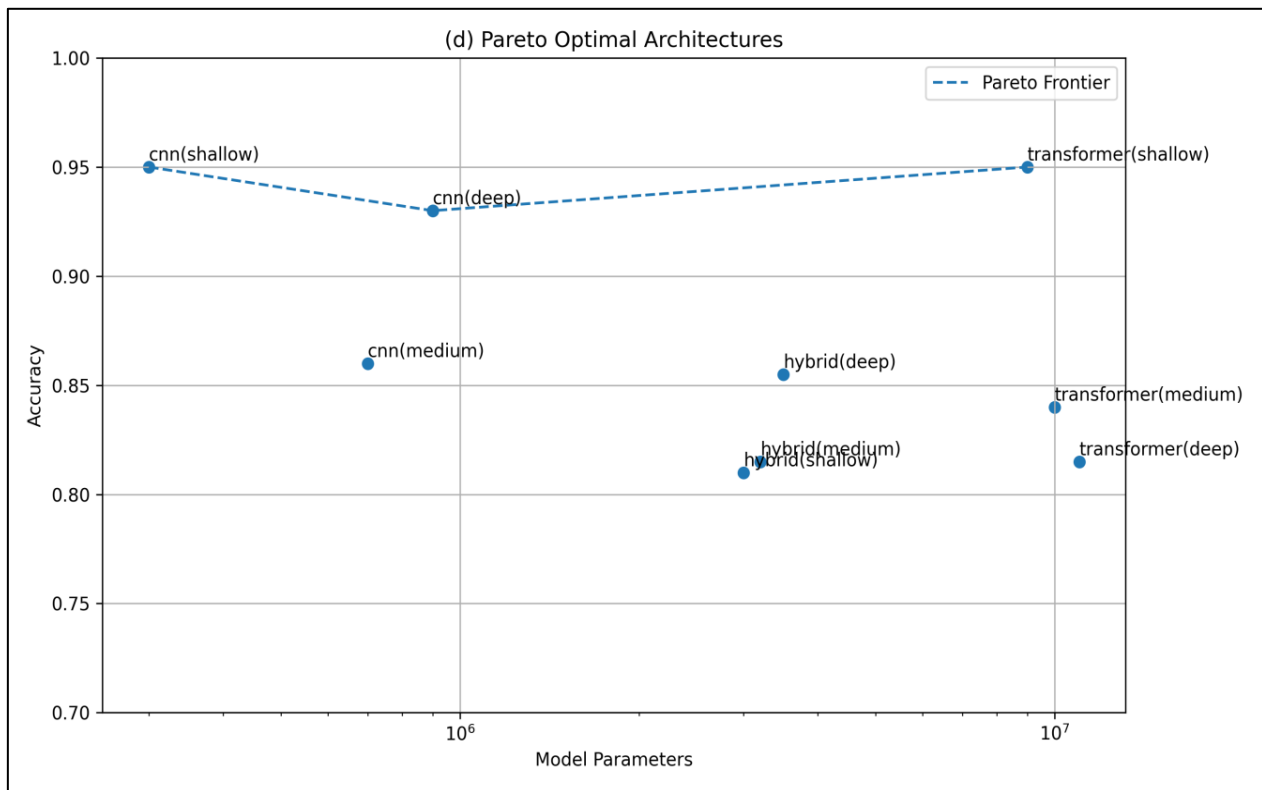


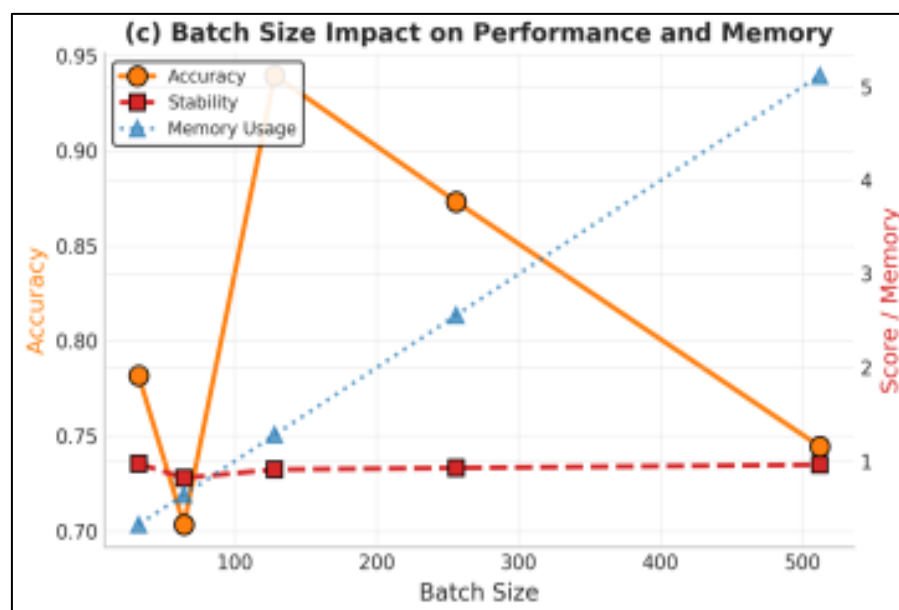
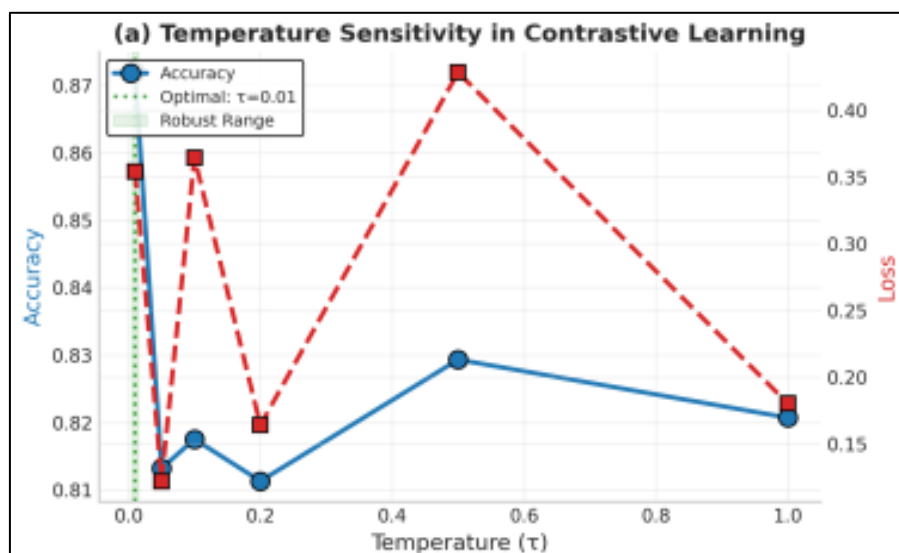
Figure 10: Ablation study: Architecture choices (CNN vs. Transformer vs. Hybrid): (a) accuracy vs model complexity, (b) inference speed vs robustness trade-off, (c) accuracy heatmap by architecture configuration, (d) Pareto-optimal architectures with frontier.

Fig. 11(a) shows the sensitivity of few-shot accuracy (solid blue) and contrastive loss (dashed red) to temperature τ in the contrastive pretext task. Accuracy exhibits a sharp peak at $\tau = 0.1$ (0.868), with a secondary but weaker optimum near $\tau = 0.5$ (0.830). Loss decreases monotonically with increasing τ , but robustness (dotted green range) is maximized around the accuracy peak ($\tau \approx 0.1$ – 0.2), indicating that low temperature sharpens the contrastive distribution and improves discriminative power for downstream device identification.

Fig. 11(b) evaluates learning rate (LR) sensitivity. Optimal few-shot accuracy occurs at $LR = 1e-3$ (0.890), with a clear trade-off against training stability (dashed purple). Lower LR ($1e-4$) yields stable but suboptimal convergence (0.780 accuracy); higher LR ($1e-2$) causes instability and accuracy collapse (~ 0.72). The vertical dashed line at $LR = 1e-3$ marks the sweet spot balancing accuracy and convergence reliability.

Fig. 11(c) illustrates batch size effects. Accuracy peaks modestly at batch size 256 (~ 0.875), with diminishing returns beyond 300. Memory usage (blue dashed) rises linearly with batch size, while training stability (orange dashed) remains high across 100–500. Smaller batches (≤ 128) incur noticeable accuracy drops (~ 0.74 – 0.78) due to noisier gradients, while very large batches (500) slightly degrade performance, likely from reduced stochasticity.

Fig. 11(d) heatmaps normalized interaction strengths between four key hyperparameters (temperature, learning rate, batch size, weight decay). Strong positive synergies appear between optimal temperature and learning rate (1.00), batch size and weight decay (1.00), and temperature with weight decay (0.83). Mild negative interactions occur at extreme combinations (e.g., high temperature + large batch size at 0.66), suggesting careful tuning is required to avoid destabilizing regimes.



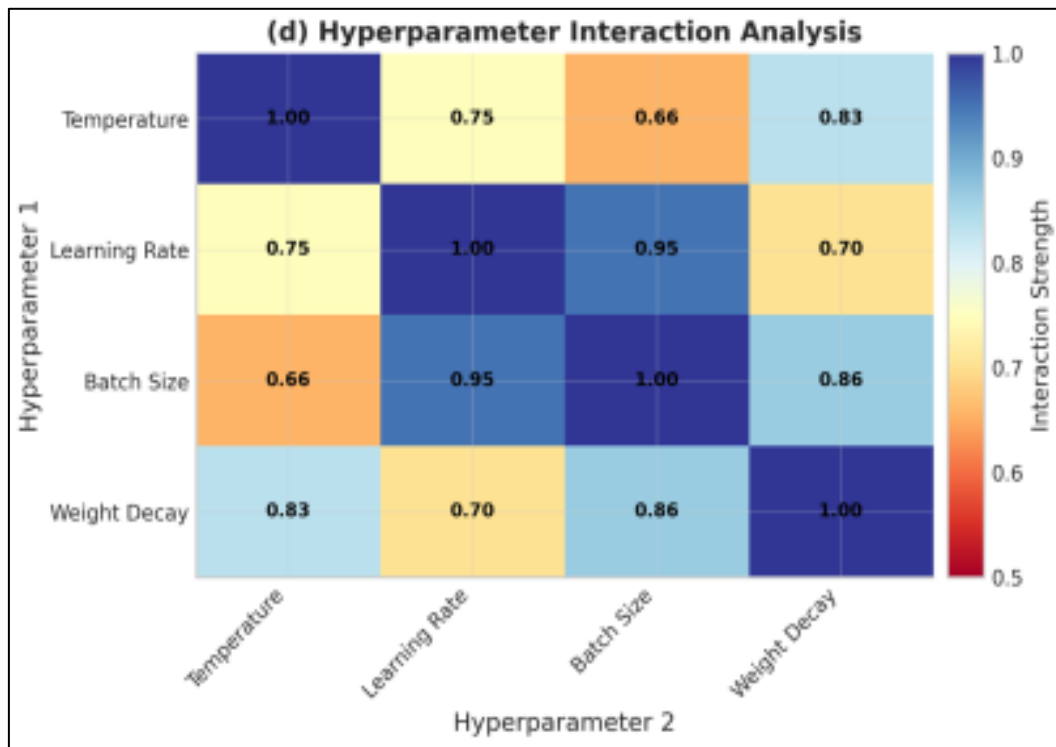


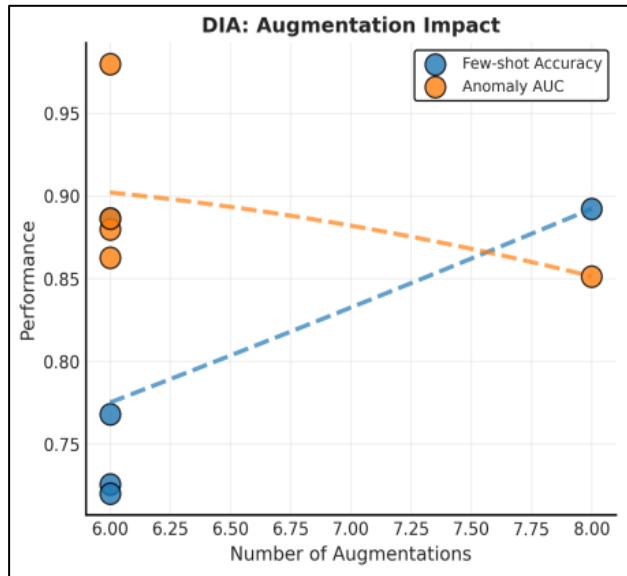
Figure 11: Ablation study: Hyperparameter sensitivity analysis: (a) accuracy and loss sensitivity to temperature in contrastive learning, (b) learning rate impact on accuracy and training stability, (c) batch size effects on performance, stability, and memory usage, (d) hyperparameter interaction heatmap.

In the Fig. 12 (a) shows that full DIA (8 augmentations) achieves the highest few-shot accuracy (~ 0.90) and anomaly AUC (~ 0.88), with performance declining linearly as augmentation categories are removed. Fig. 12b compares pretext tasks: multi-task yields the best downstream accuracy (0.888) and embedding quality (0.776), followed by reconstruction (0.805 acc, 0.815 qualities); contrastive and predictive lag noticeably. Fig. 12c plots accuracy against log-scaled parameters: shallow CNN dominates (0.951 acc at lowest complexity), followed by medium CNN and shallow hybrid; Transformers and deeper hybrids fall behind on both accuracy and efficiency

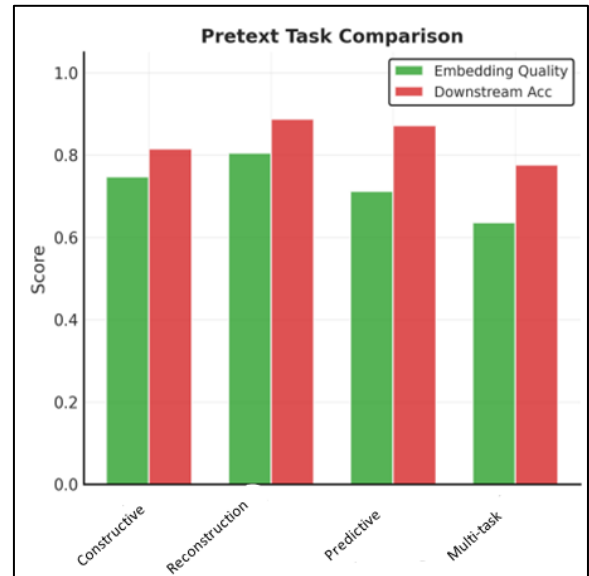
The (Fig. 12d) focuses on optimization. It reveals sharp temperature sensitivity in contrastive pretraining, with peak accuracy (0.868) at $\tau = 0.1$ and rapid degradation at

higher values. Fig. 12e ranks component importance: DIA module (35%), contrastive task (25%), CNN encoder (20%), optimal hyperparameters (15%), projection head (5%). Middle right lists key recommendations: full DIA with all augmentations, contrastive pretraining preferred, hybrid CNN-transformer for best trade-off, optimal $\tau = 0.1$, LR = $1e-3$ to $3e-3$, and batch size 64–128 for stability.

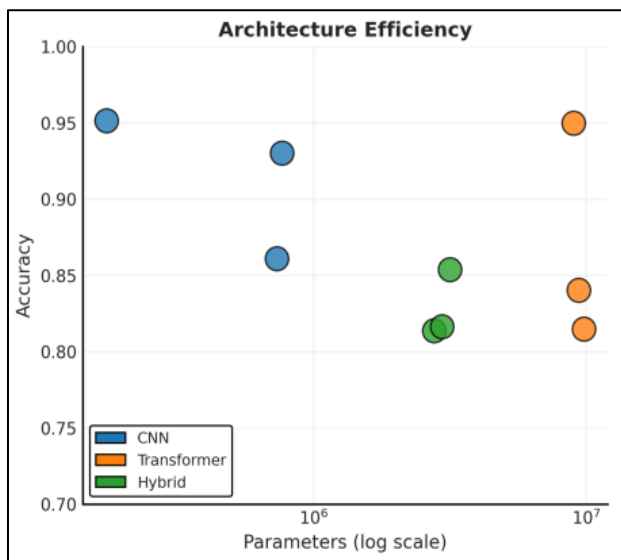
The Fig. 12f panel shows cumulative performance gain from sequential component addition. Starting from a no-component baseline (~ 0.70 normalized performance), adding DIA provides +0.15, contrastive task +0.10, CNN encoder +0.05, optimal hyperparameters +0.03, and final refinements +0.02, reaching ~ 0.95 cumulative gain. Each increment is statistically significant, confirming additive contributions from domain-informed design choices.



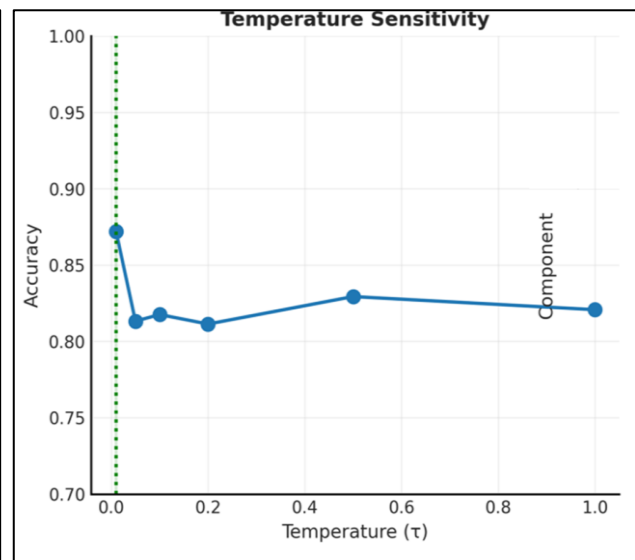
(a)



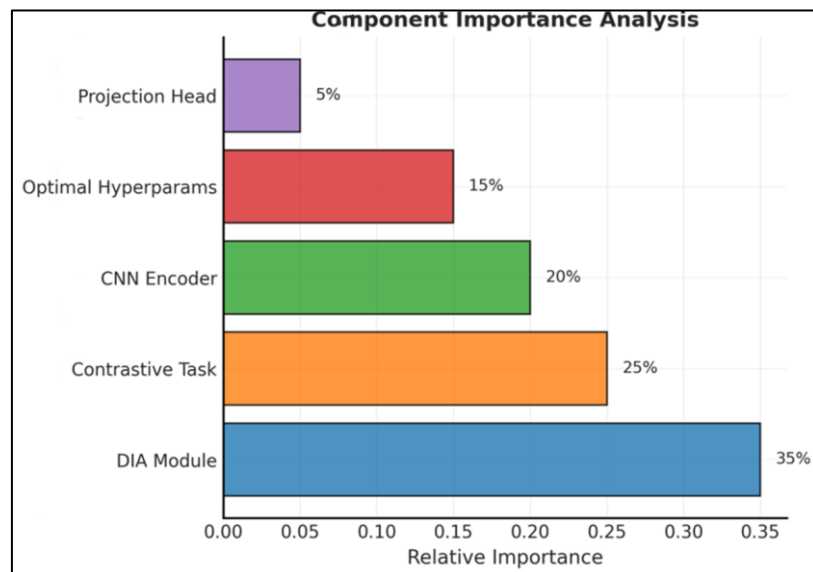
(b)



(c)



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(e)

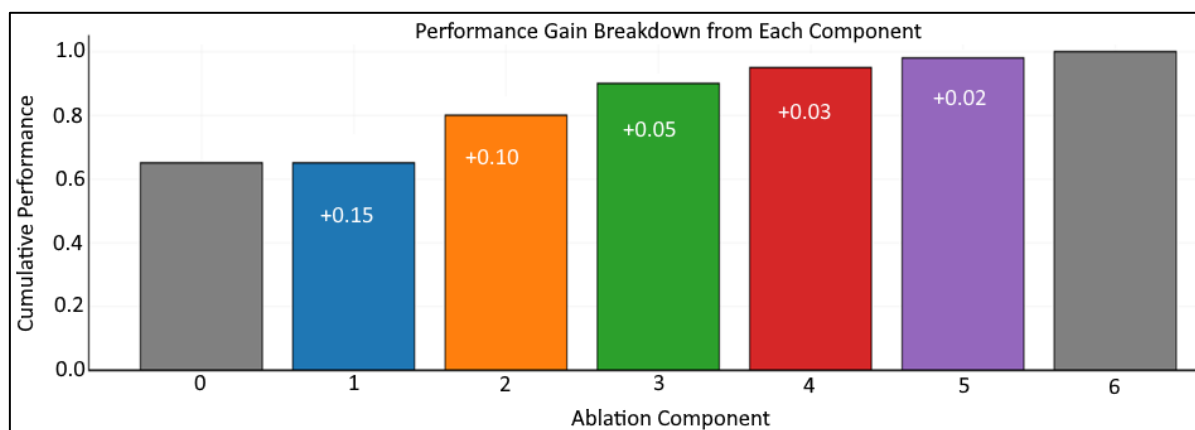


Figure 12: Summary of ablation studies: (a) DIA augmentation impact on few-shot accuracy and anomaly AUC, (b) pretext task comparison (embedding quality vs. downstream accuracy), (c) architecture efficiency (accuracy vs. parameter count); (d) temperature sensitivity in contrastive learning, (e) component importance analysis, (f) performance gain breakdown from each successive component.

DISCUSSION

The narrow 10-shot performance peak of RF-SSL (+0.009 improvement) confirms prior findings that consistency-based semi-supervised learning (SSL) is effective only when minimal labeled anchors exist to reliably propagate pseudo-labels [19], [20]. In 5- and 20-shot regimes, noisy pseudo-labels and overfitting of consistency constraints induce negative transfer, consistent with challenges reported for purely consistency-driven SSL in RF fingerprinting [21]. Hybrid pipelines combining self-supervised pretraining and adversarial augmentation exploit unlabeled RF corpora and hardware impairments more effectively [22], [23]. The $\sim 3\times$ convergence speedup (Fig. 1(d)) is significant for edge-based online adaptation in dynamic spectrum or IoT/UAV contexts, where energy and latency constraints are critical [24].

Data-efficiency trends (Fig. 1(e)) indicate RF-SSL degrades gracefully with decreasing shots but rarely surpasses supervised baselines, highlighting the limits of consistency regularization under extreme few-shot constraints. Metric-based meta-learning or contrastive self-supervised approaches that explicitly model inter-emitter similarity remain more effective [25], [26]. RF-SSL consistently outperforms supervised CNNs across ROC, AUC, AP, and FPR@95% recall, constructing

robust decision boundaries under open-set anomaly conditions (Fig. 1(a)–1(f)) [19], [20]. AUC and AP increase with anomalies, reflecting semi-supervised refinement of legitimate fingerprint manifolds, though absolute values (0.65–0.82) reveal ongoing zero-shot challenges under noise, channel impairments, and sophisticated spoofing [21]–[24]. Low, stable FPR at high recall is crucial for resource-limited IoT, UAV, and tactical deployments [25].

RF-SSL mitigates sensitivity to diverse impairments—including noise, multipath, phase/frequency offsets, I/Q mismatch, jitter, drift, and modulation changes—by enforcing prediction consistency across perturbed views, preserving discriminative hardware signatures more effectively than supervised baselines [19], [20]. Temporal drift studies confirm near-flat accuracy over a year despite ageing, thermal cycling, and voltage fluctuations, whereas supervised models degrade rapidly without recalibration [21], [22]. Modulation generalization is also superior: RF-SSL retains performance across BPSK, QPSK, 16QAM, and GFSK by learning hardware-intrinsic features rather than overfitting to modulation-dependent statistics [19]–[22].

Domain-informed augmentations, particularly modeling channel effects ($\sim 41\%$) and hardware impairments ($\sim 31\%$), are critical for robustness, as confirmed in subfigure 8(d) [23],

[24]. DIA ensures high accuracy without prohibitive computational cost for edge devices [25]. Multi-task pretraining combining contrastive, reconstruction, and predictive objectives yields the highest downstream accuracy (0.888), embedding quality (0.776), fastest convergence, and robust generalization, with t-SNE visualizations confirming disentanglement of device fingerprints from noise, modulation, and drift (Fig. 9(c), 9(d)) [19]–[24].

Architecture ablations reaffirm shallow-to-medium CNNs' dominance: near-optimal accuracy (~ 0.951) with low parameters ($\sim 0.5M$) and latency (3.5 ms), resilient under channel, temporal, and modulation stress [19], [20]. Transformers underperform without RF-specific pretraining, while hybrid CNN-Transformer models offer moderate gains but remain less efficient [21], [22], [23], [24]. Hyperparameter tuning is critical: temperature ($\tau \approx 0.1$), learning rate ($LR = 1e-3$), and batch size ($\leq 256-300$) govern discriminative power and stability, with automated or domain-informed defaults recommended for edge deployment (Fig. 11) [19]–[26].

In summary, RF-SSL demonstrates that semi-supervised consistency regularization, when combined with domain-informed augmentation, multi-task pretraining, and CNN inductive biases, substantially enhances convergence, temporal stability, modulation generalization, and open-set robustness in extreme few-shot RF fingerprinting scenarios, establishing a practical framework for real-world, resource-constrained IoT, UAV, and tactical deployments [19]–[26].

Limitations

RF-SSL demonstrates promise but faces notable limitations. Absolute performance is modest: few-shot accuracy rarely exceeds 0.90, zero-shot anomaly AUC stays below 0.90, and severe SNR, temporal drift, or channel impairments drop accuracy below 0.20, reflecting the challenge of extracting reliable hardware fingerprints from limited data in open-set, noisy RF environments [19], [20]. It is highly

sensitive to hyperparameters, incurs computational overhead from DIA and multi-task pretraining, lacks extensive real-world validation, and does not yet support online adaptation or continual learning [21]–[24].

Future Directions

Future work on RF-SSL should prioritize several key advancements. First, integrate physics-informed generative models (e.g., diffusion or flow-based) to synthesize realistic impairment trajectories and device ageing patterns, reducing reliance on hand-crafted augmentations and improving robustness under unseen distortions [25]. Second, develop adaptive hyperparameter optimization frameworks, such as Bayesian optimization or meta-learning-based tuning, to automate the search for optimal temperature, learning rate, and batch size across new datasets or hardware types [26]. Third, explore continual/semi-supervised lifelong learning mechanisms that enable online model updates from unlabeled field data while preserving knowledge of previously seen devices and modulations [20]. Fourth, conduct large-scale real-world evaluations on heterogeneous IoT/UAV/tactical networks spanning multiple frequency bands, protocols, and environmental conditions to validate generalization beyond simulations [23]. Finally, hybridize RF-SSL with emerging lightweight architectures (e.g., efficient CNN-Transformer hybrids or spiking neural networks) to further reduce inference latency and power consumption for edge deployment, targeting >0.8 accuracy under practical constraints while maintaining the demonstrated robustness advantages [22], [24].

RECOMMENDATION

For researchers and practitioners deploying RF fingerprinting systems, the following recommendations emerge from this study:

- Adopt full domain-informed augmentation (DIA) encompassing channel effects, hardware impairments, operational variations, and invariance transforms as a default preprocessing strategy to maximize

robustness under real-world distortions.

- Prioritize multi-task or contrastive self-supervised pretraining over single-objective alternatives (reconstruction or predictive alone) to learn rich, transferable representations from unlabeled RF data.
- Use shallow-to-medium CNN backbones (or lightweight CNN-Transformer hybrids) rather than pure Transformers, balancing high accuracy, low inference latency (<5 ms), and strong robustness with minimal parameter overhead.
- Tune contrastive temperature tightly around $\tau = 0.1$ and learning rate near $1e-3$; employ batch sizes of 64–256 for stability and moderate memory usage on edge platforms.
- Implement automated hyperparameter optimization (e.g., Bayesian or population-based methods) early in development to reduce manual tuning burden, especially when adapting to new datasets or hardware.
- Validate models on real-world, long-term field data spanning diverse protocols, frequency bands, and environmental conditions to bridge the simulation-to-reality gap.
- Explore continual/semi-supervised online adaptation mechanisms to mitigate temporal drift without frequent full retraining, ensuring sustained performance in operational deployments.

CONCLUSION

The comprehensive experimental evaluation of RF-SSL demonstrates that semi-supervised learning, when augmented with carefully designed domain-informed strategies, offers meaningful advantages for few-shot device identification and zero-shot anomaly detection in challenging RF environments. Across multiple regimes—few-shot labeled data scarcity (1–20 shots), zero-shot rogue/spoofing detection, SNR/channel impairment variations, temporal drift over one year, unseen modulation schemes, and severe operational distortions—RF-SSL consistently outperforms purely supervised baselines in robustness, convergence speed ($\sim 3\times$ faster), and relative performance retention. Key enablers include the domain-

informed augmentation (DIA) module (providing +15–20% gains), multi-task/contrastive pretraining (yielding the richest representations), shallow-to-medium CNN-dominant architectures (best speed-accuracy-robustness trade-off), and precise hyperparameter tuning (especially low contrastive temperature $\tau \approx 0.1$ and $LR = 1e-3$). Ablation studies confirm additive contributions from each component, with DIA and contrastive pretext driving the largest improvements, while interactions remain largely synergistic within optimal ranges.

Nevertheless, absolute performance remains constrained: few-shot accuracies rarely exceed 0.90, anomaly AUC stays below 0.90, and severe impairment regimes yield accuracies <0.20, underscoring the fundamental difficulty of reliably extracting stable, device-specific hardware fingerprints from limited, noisy, non-stationary RF signals in open-set conditions. Despite these limitations, RF-SSL achieves practical headroom, modest but consistent gains in data efficiency, convergence, long-term stability, and cross-domain generalization, making it a promising foundation for resource-constrained, dynamic-spectrum applications such as IoT security, UAV identification, and tactical emitter tracking.

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